

*Pf*MORC protein regulates chromatin accessibility and transcriptional repression in the human malaria parasite, *Plasmodium falciparum*

Reviewed Preprint

v2 • October 18, 2024

Revised by authors

Reviewed Preprint

v1 • November 20, 2023

Zeinab M Chahine, Mohit Gupta, Todd Lenz, Thomas Hollin, Steven Abel, Charles AS Banks, Anita Saraf, Jacques Prudhomme, Suhani Bhanvadia, Laurence A Florens, Karine G Le Roch 

Department of Molecular, Cell and Systems Biology, University of California Riverside, Riverside, USA • Stowers Institute for Medical Research, Kansas City, USA

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Abstract

The environmental challenges the human malaria parasite, *Plasmodium falciparum*, faces during its progression into its various lifecycle stages warrant the use of effective and highly regulated access to chromatin for transcriptional regulation. Microrchidia (MORC) proteins have been implicated in DNA compaction and gene silencing across plant and animal kingdoms. Accumulating evidence has shed light into the role MORC protein plays as a transcriptional switch in apicomplexan parasites. In this study, using CRISPR/Cas9 genome editing tool along with complementary molecular and genomics approaches, we demonstrate that *Pf*MORC not only modulates chromatin structure and heterochromatin formation throughout the parasite erythrocytic cycle, but is also essential to the parasite survival. Chromatin immunoprecipitation followed by deep sequencing (ChIP-seq) experiments suggest that *Pf*MORC binds to not only sub-telomeric regions and genes involved in antigenic variation but may also play a role in modulating stage transition. Protein knockdown experiments followed by chromatin conformation capture (Hi-C) studies indicate that downregulation of *Pf*MORC impairs key histone marks and induces the collapse of the parasite heterochromatin structure leading to its death. All together these findings confirm that *Pf*MORC plays a crucial role in chromatin structure and gene regulation, validating this factor as a strong candidate for novel antimalarial strategies.

eLife Assessment

This **important** study examines the role of Microrchidia (MORC) proteins in the human malaria parasite *Plasmodium falciparum*. **Solid** experimental results, including genome editing and chromatin profiling methods (ChIP-seq and Hi-C), provide a comprehensive picture of the critical role MORC plays in shaping parasite chromatin. Depletion of MORC results in a lethal collapse of heterochromatin and parasite death, nominating the factor as a new target of antimalarial therapies.

<https://doi.org/10.7554/eLife.92499.2.sa2>

Introduction

Malaria is a mosquito-borne infectious disease that is caused by protozoan parasites of the genus *Plasmodium*. Among the five human-infecting species, *Plasmodium falciparum* is the deadliest, with over 619,000 deaths in 2023(1). To adapt to extreme environmental challenges, *P. falciparum* possesses unique strategies that direct the tightly coordinated changes in gene expression and control transcriptional switching in genes encoded by multigene families to ensure antigenic variation and immune evasion. Changes in gene expression throughout the parasite life cycle are controlled by a surprisingly low repertoire of transcription factors (TFs) encoded in the *Plasmodium* genome (2–9). The 27 apicomplexan APETALA2 (ApiAP2) DNA-binding proteins are the only well documented parasite TFs known to contribute to the modulation of gene expression throughout various stages of the parasite's development(3, 5–8, 10–13). Since the discovery of the AP2 gene family(3), evidence has alluded to their role as master regulators of gene expression throughout transitory phases of the parasite life cycle. The most well documented are the gametocyte specific TFs (AP2-G, AP2-G2)(7, 14) with subsequent discovery of those associated with transition to sexual differentiation(13) as well as sporozoite (AP2-SP) and liver stages (AP2-L) (6, 10). The *Plasmodium* genome, however, encompasses well over 5,000 protein-coding genes suggesting there are most likely other molecular components responsible for gene expression. It is believed that, to offset the relatively low TF range, *Plasmodium* has evolved additional mechanisms regulating gene expression including mechanisms that use epigenetics factors, RNA binding proteins or regulate chromatin structure (15–20). All together these components work in combination to regulate the dynamic organization of DNA and the cascade of gene expression required for the parasite life cycle progression(21–28). Despite their importance, the identification and functional characterization of regulatory players controlling chromatin remains a challenge in this intractable organism. However, with the advent of sensitive technologies capable of capturing important chromatin associated regulatory complexes such as chromatin immunoprecipitation (ChIP), chromatin conformation capture Hi-C technologies, and chromatin enrichment for proteomics (ChEP), we can now solve important facets of molecular components controlling epigenetics and chromatin structure.

Microrchidia (MORC) belong to a highly conserved nuclear protein superfamily with widespread domain architectures that link MORCs with signaling-dependent chromatin remodeling and epigenetic regulation across plant and animal kingdoms, including the apicomplexan parasites(29–32). In all organisms, MORC contains several domains that form a catalytically active ATPase. Apicomplexan MORC ATPases are encircled by Kelch-repeat β -propellers as well as a CW-type zinc finger domain functioning as a histone reader(33). In higher eukaryotes, MORCs were first identified as epigenetic regulators and chromatin remodelers in germ cell development. Currently, these proteins are shown to be involved in various human diseases including cancers and are expected to serve as important biomarkers for diagnosis and treatment(34). In the apicomplexan parasite, *Toxoplasma gondii*, TgMORC was shown to recruit the histone deacetylase HDAC3 to particular genome loci to regulate chromatin accessibility, restricting sexual commitment (33, 35). TgMORC-depleted cells also resulted in change in gene expression with up regulation of secondary AP2 factors and a significant shift from asexual to sexual differentiation(32, 33, 36–38). Having multiple homologs with *T. gondii*, *Plasmodium* AP2 protein conservation is primarily restricted to their AP2 DNA-binding domains (39, 40). Recently, ChEP experiments done in *P. falciparum* identified PfMORC as one of the highest enriched chromatin-bound proteins at different stages of the parasite intraerythrocytic development cycle (IDC)(28). PfMORC was also detected at a relatively high level throughout the parasite life cycle, including sporozoites and liver stages(33, 41), and has been identified in several protein pull down experiments as targeting both AP2 TFs and epigenetic factors(32, 33, 38, 42–44). Immunoprecipitation experiments demonstrated that PfMORC seems to interact with AP2-

G2(32 [4](#)), a TF that plays a critical role in the maturation of gametocytes. However, the genome wide distribution and exact function of this protein throughout the parasite life cycle remained elusive.

In this study, we apply CRISPR/Cas9 genomic editing technologies to determine the function, genome distribution, and indispensability of *PfMORC* throughout the parasite's IDC. Immunoprecipitation of an HA-tagged parasite line validates the role of *PfMORC* in heterochromatin structure maintenance. Using downregulation of *PfMORC* induced through TetR-DOZI system, we demonstrate the functional significance of *PfMORC* in heterochromatin stability and gene repression. Immunofluorescence based assays and ChIP-seq experiments show that *PfMORC* localizes to heterochromatin clusters at or near *var* genes with significant overlap with well-known signatures of the parasite heterochromatin post-translational modification (PTM) H3K9-trimethylation (H3K9me3) marks. When *PfMORC* was downregulated, level of H3K9me3 was detected at a lower level, demonstrating a possible role of this protein in epigenetic regulation and gene repression. Finally, Hi-C analyses demonstrate that downregulation of *PfMORC* results in significant dysregulation of chromatin architectural stability resulting in parasite death. All together our work provides significant insight into the role of *PfMORC* in the maintenance of the pathogens' epigenetics and chromosomal architectural integrity and validates this protein as a promising target for novel therapeutic interventions.

Results

Generation of *PfMORC*-HA transgenic line shows nuclear localization in heterochromatin clusters

To characterize the role of *PfMORC* in *P. falciparum*, we applied CRISPR/Cas9 genome editing tool to add a 3X-HA tag at the C-terminal coding region of the *PfMORC* locus (PF3D7_1468100) in an NF54 line(45 [4](#)) (Fig. 1a [4](#), Sup. data 1). Recovered transgenic parasites were cloned using limiting dilution and the correct incorporation of the tag into the parasite genome was validated via PCR and whole genome sequencing (WGS) (Fig. 1b [4](#), Sup. data 2). WGS analysis confirmed the presence of the HA tag at the expected locus with no obvious off-target effect but uncovered an additional nonsense mutation in the *gametocyte development protein 1* (*gdv1*) gene (PF3D7_0935400) at amino acid position 561 out of 599 total. Mutations in *gdv1* have been detected in the past and seem to emerge relatively often, indicating a fitness benefit in the *in vitro* culture system used in the laboratory(8 [4](#)). The involvement of GDV1 as essential in sexual commitment is well substantiated and further study involving the role of *PfMORC* in sexual commitment could not be fully addressed(7 [4](#), 8 [4](#), 14 [4](#), 46 [4](#)–48 [4](#))(Sup. data 2). Despite this finding, we were able to design experiments to determine the role of *PfMORC* throughout the IDC as no other major detrimental mutations were detected in the genome. We first validated *PfMORC* protein expression through Western blot analysis (Fig. S1). Regardless of potential protein degradation, results showed an expected band size within the ~290 kDa range that was absent in NF54 control compared to our tagged line (Fig. S1). Immunofluorescence (IFA) analysis of intracellular parasites revealed that *PfMORC* is localized in the nucleus throughout the parasite IDC in punctate patterns (Fig. 1c [4](#)). A single foci per parasite was detected in the nuclear periphery at the ring stage. An increased number of foci with a more diffuse signal could be detected as the number of DNA copies increased at the schizont stage. Moreover, punctuates were shown to have strong colocalization signals with histone H3K9me3 marks present at the different parasite developmental stages analyzed. This was most evident during the early ring stage of the parasite IDC where the chromatin organization is more compact with a single DNA copy. Altogether, our findings indicated that *PfMORC* is a true nuclear protein that is most likely associated with *P. falciparum* heterochromatin cluster(s).

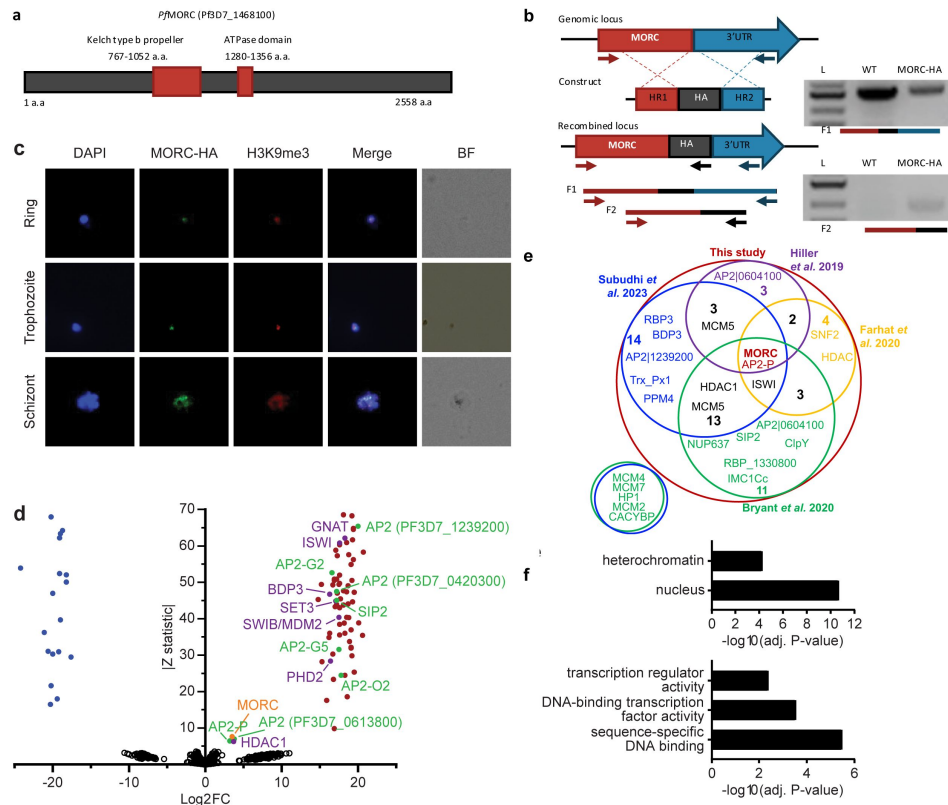


Fig. 1.

***PfmMORC*-HA is associated with heterochromatin.**

(a) Illustration of the *PfmMORC* containing domains including Kelch type b propeller and ATPase domains using InterProScan. (b) Design strategy applied for *PfmMORC* C-terminal HA tagging. PCR amplification of the genomic C-terminus end of *Pfmorc* region extending towards the 3'UTR (F1) as well as extension from the C-terminus towards the HA flanking sequence (F2) verifies the correct insertion site. NF54 genomic DNA was used as negative control. (c) IFA experiment: *PfmMORC* foci²² expressing co-localization with H3K9me3 marks (red). Cell nuclei are stained with DAPI (blue). BF: brightfield (d) Protein immunoprecipitation: Significance plot representing *PfmMORC* interactome recovered through immunoprecipitation followed by mass spectrometry (IP-MS). Graph list MORC (orange) bindings partners of highest affinity associated with TF regulation^{22 22} and chromatin remodelers, erasers, and writers (purple). Proteins enriched in the *PfmMORC*-HA samples compared with controls were filtered with $\log_2 FC \geq 2$ and Z statistic > 5 . (e) Venn diagram representing overlapping proteins identified among five publications. Values represent the total number of significant proteins identified as overlapping between two subsets. (f) Gene ontology enrichment analysis of the significantly enriched proteins. The top 2 terms of Cellular Component (top) and top 3 terms of Molecular Function (bottom) are represented as $-\log_{10}$ (adjusted P-value) (Fisher's exact test with Bonferroni adjustment).

***Pf*MORC interacting partners include proteins involved in heterochromatin maintenance, transcription regulation and chromatin remodeling**

To confirm the association of *Pf*MORC with proteins involved in heterochromatin cluster(s), we performed immunoprecipitation (IP) followed by mass-spectrometry (MS) analysis using mature stages of *Pf*MORC-HA and parental line as control. MORC was detected with highest peptides (97 and 113) and spectra (1041 and 1177) counts compared to control conditions (5 and 7 peptides; 16 and 43 spectra) confirming the efficiency of our pull-down. However, considering the relatively large size of the MORC protein (295kDa) and its weak detection in the control, the log₂ FC and Z-statistic after normalization are minimal when compared to smaller proteins that were not identified in the control samples. To define a set of *Pf*MORC associated proteins, we used the QPROT statistical framework(49) to compare proteins detected in *Pf*MORC samples and controls. We identified 73 *Pf*MORC associated proteins (log₂ FC ≥ 2 and Z statistic > 5) (Fig. 1d, Sup. data 3). Additional chromatin remodeling proteins were detected such as ISWI chromatin-remodeling complex ATPase (PF3D7_0624600) and SWIB/MDM2 domain-containing protein (PF3D7_0611400), as well as the epigenetic readers BDP3 (PF3D7_0110500) and PHD2 (PF3D7_1433400)(50). We also detected a significant enrichment of histone erasers and writers such as a N-acetyltransferase (PF3D7_1020700), HDAC1 (PF3D7_0925700), and SET3 (PF3D7_0827800). Several of which have been detected to interact with MORC in previous studies including the AP2-G5, HDAC1, ELM2 and ApiAP2 proteins(33, 42–44) (Fig. 1e). Notably, we observed that many of the detected proteins were enriched at levels comparable to those of the bait protein such as AP2-P and HDAC1. This may be a result of proteins forming complexes in a one-to-one ratio, leading to similar enrichment levels. Notably, two of these three proteins have been reported to interact with MORC in several studies, further supporting a strong interaction between them. Gene Ontology (GO) enrichment analysis revealed that these proteins are nuclear and are associated with heterochromatin, DNA binding, and transcription factor activity (Fig. 1f). Detailed analysis showed the detection of SIP2, involved in heterochromatin formation and chromosome end regulation(51), and 8 AP2 transcription factors, including AP2-G2, AP2-O2, AP2-G5. Among the other AP2s detected, 5 were previously identified by ChIP-seq as *Plasmodium falciparum* AP2 Heterochromatin-associated Factors (*Pf*AP2-HFs)(52). Our findings, corroborated with several published works, indicate that *Pf*MORC interacts with multiple ApiAP2 TFs, chromatin remodelers, and epigenetic players associated with heterochromatin regions.

Genome-wide distribution of *Pf*MORC displays functional roles in stage-specific gene silencing and antigenic variation

Given the multifunctional role of MORC proteins throughout apicomplexan parasites, we further explored the association of *Pf*MORC with chromatin accessibility and stage-specific gene regulation by assessing the genome-wide distribution of *Pf*MORC using chromatin immunoprecipitation followed by deep sequencing (ChIP-seq) in duplicate at the ring, trophozoite and schizont stages. Inspection of *Pf*MORC binding sites across each chromosome revealed a strong signal for telomeric and subtelomeric regions of the parasite genome, as well as the internal clusters of genes responsible for antigenic variation such as *var* and *rifin* (Fig. 2a-c, Sup. data 4). We found that while only 47% of reads mapped to antigenic genes in the ring stage, 95% of reads mapped to the same genes during the trophozoite stage and 82% during the schizont stage. Although a vast majority of the genome is actively transcribed during the trophozoite stage, these antigenic gene families are under tight regulatory control to ensure mutually exclusive expression and participate in immune evasion(53, 54). This could explain the disparity in the level of *Pf*MORC binding of the coding regions between the ring and trophozoite stages (Fig. 2c, Fig. S2b).

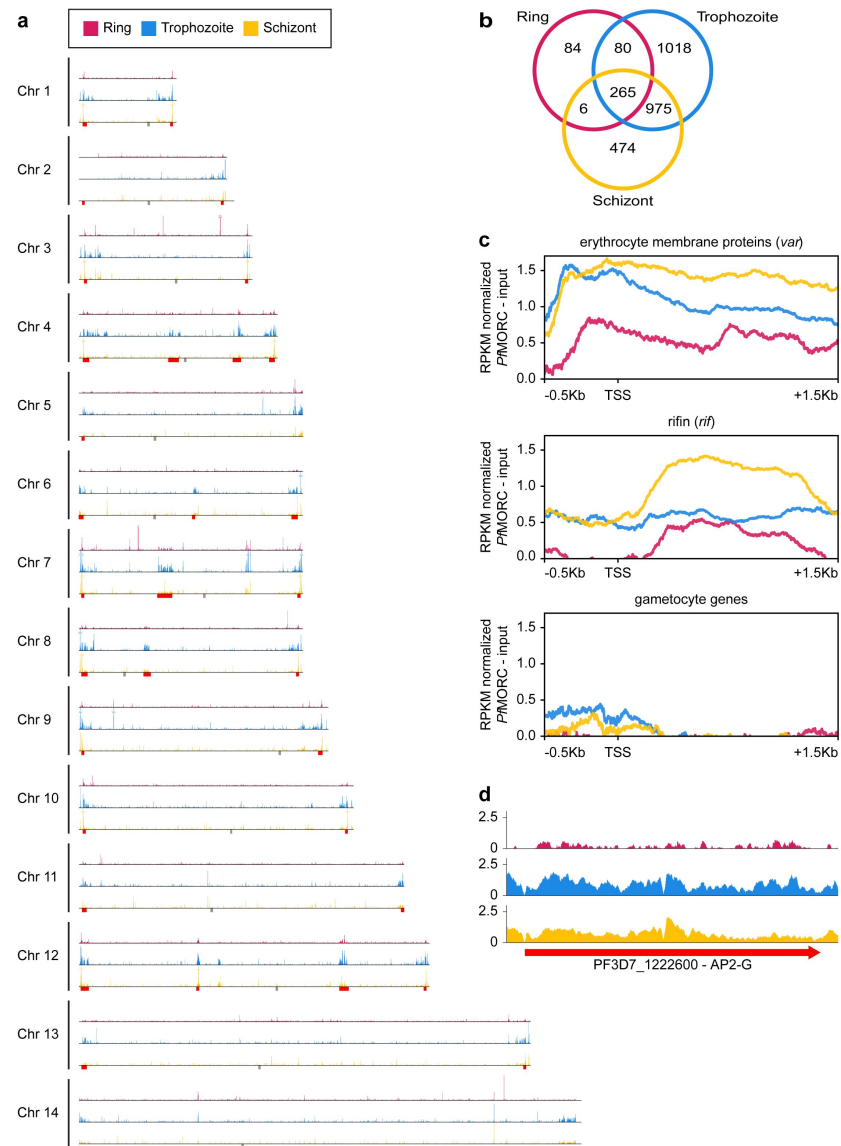


Fig. 2.

Genome wide distribution of *PfMORC* proteins.

(a) Chromosome distribution plots of *PfMORC* binding showing a predisposition for subtelomeric and internal *var* gene regions (red). Each track is input subtracted, and per-million read count normalized before normalizing the track height to allow for direct comparison between stages. Grey boxes indicate the position of centromeres. (b) Overlap of called peaks between time points. (c) Profile plots showing *PfMORC* coverage from 0.5 kb 5' of the transcription start site (TSS) to 1.5 kb 3' of the TSS in the ring, trophozoite, and schizont stages. Each plot includes per-million read count normalized coverage at 1 bp resolution for all genes within the *var* and *rifin* gene families as well as gametocyte-specific genes. (d) *PfMORC* coverage of the gametocyte-specific transcription factor *ap2-g*.

We also evaluated *Pf*MORC binding of stage-specific gene families including gametocyte-related genes, and merozoite surface proteins (*msp*)(55). In trophozoite and schizont stages, gametocyte-associated genes contain a mean of <0.5 RPKM normalized reads per nucleotide of *Pf*MORC binding within their promoter region, whereas antigenic gene families such as *var* and *rifin* contain ~1.5 and 0.5 normalized reads, respectively (Fig. 2b). The difference is even greater within the gene body with gametocyte genes displaying almost no reads mapped more than 200 bp downstream of the TSS and antigenic gene families containing 0.5-1.5 RPKM normalized reads. However, the gametocyte-specific transcription factor AP2-G, known to be repressed during the IDC and required for sexual commitment, deviates from this trend and contains similar levels of *Pf*MORC binding to *var* genes (Fig. 2c and S2c, d). These results indicate a major role of *Pf*MORC in controlling AP2-G and sexual differentiation. For some of the *msp* genes that are usually expressed later during the IDC to prepare the parasite for egress and invasion of new erythrocytes, *Pf*MORC binding was detected in the gene bodies at the trophozoite stage. We also detected a small switch in *Pf*MORC binding sites, moving from their gene bodies to their intergenic regions at the schizont stage (Fig. S2d). *Pf*MORC most likely moved away from the gene body to the regulatory regions surrounding the transcription start site (TSS) to guide RNA Polymerase and transcription factors and aid in activating expression of genes at the schizont stage that are crucial for egress and invasion. These results provide strong evidence for the direct effects that *Pf*MORC binding has on tightly controlled antigenic and stage-specific genes including crucial invasion and gametocyte genes.

***Pf*MORC is essential for *P. falciparum* survival**

We next sought to confirm the functional relevance of *Pf*MORC protein in parasite survival. While partial *Pf*MORC knockdown using the glmS-ribozyme system had previously shown to have no significant effect on the parasite survival(41), another study using transposon mutagenesis (*piggyBac*) identified *Pf*MORC as likely essential(56). To resolve these conflicting results, we applied a complementary approach using the CRISPR/Cas9 gene editing strategy to incorporate an inducible TetR-DOZI system to knockdown (KD) *Pf*MORC(57) through administration of anhydrotetracycline (aTC) (Fig. 3a)(45, 58). The protein was also modified to include a C-terminal 3x-HA tag. Parental and transgenic clones were validated via PCR (Fig. 3b) and WGS to confirm the correct insertion of the inducible system (Sup. data 2) as well as the absence of major mutation that could explain some of the phenotypes observed. While results from our WGS validated our editing strategy without the detection of any obvious deleterious off-target effect, we identified a nonsense mutation in AP2-G gene (PF3D7_1222600) explaining our inability to obtain mature gametocytes. Protein expression and successful KD was confirmed via western blot with a decrease of ~ 58% and ~ 88.2% in *Pf*MORC expression at 24 hpi and 36 hpi respectively compared to their aTC supplemented conditions (Fig. S3). Downregulation of *Pf*MORC was detected significantly above the level observed by Singh and colleagues(41).

We then performed a phenotypic assay on (+/-) aTC *Pf*MORC-HA-TetR-DOZI parasite cultures. The assay was conducted in replicates of synchronized cultures that were then split into (+) aTC and (-) aTC conditions either at the ring or trophozoite stages on *Pf*MORC-HA-TetR-DOZI clones and WT control lines (Fig. 3c-f, Sup. data 5). Sequential morphological screens were performed through Giemsa-stained smears and monitored by microscope imaging. As opposed to what was observed previously with the glmS-ribozyme system(41), aTC removal at the ring stage induced clear signs of stress and cell cycle arrest in mid-trophozoite and schizont stages of the first intraerythrocytic cycle (Fig. 3c-d) compared to (+) aTC *Pf*MORC and WT parasites. Our data showed an ~53%, 77%, and 84% (n=3, p<0.0001) drop in parasitemia at 48-, 72- and 120-hours post invasion (hpi), respectively, when compared to control conditions. aTC supplemented and WT cultures showed unperturbed cell cycle progression, reinvasion, and morphological development. Interestingly, when aTC was withheld at the late trophozoite stages (24 to 30 hpi), parasites could complete the first cycle and successfully invade into new red blood cells (RBCs) (Fig. 3e-f). The evident hindered phenotypic response may either be associated with the delay time for the protein to be completely depleted or may suggest a more dispensable role of *Pf*MORC in the schizont stage.

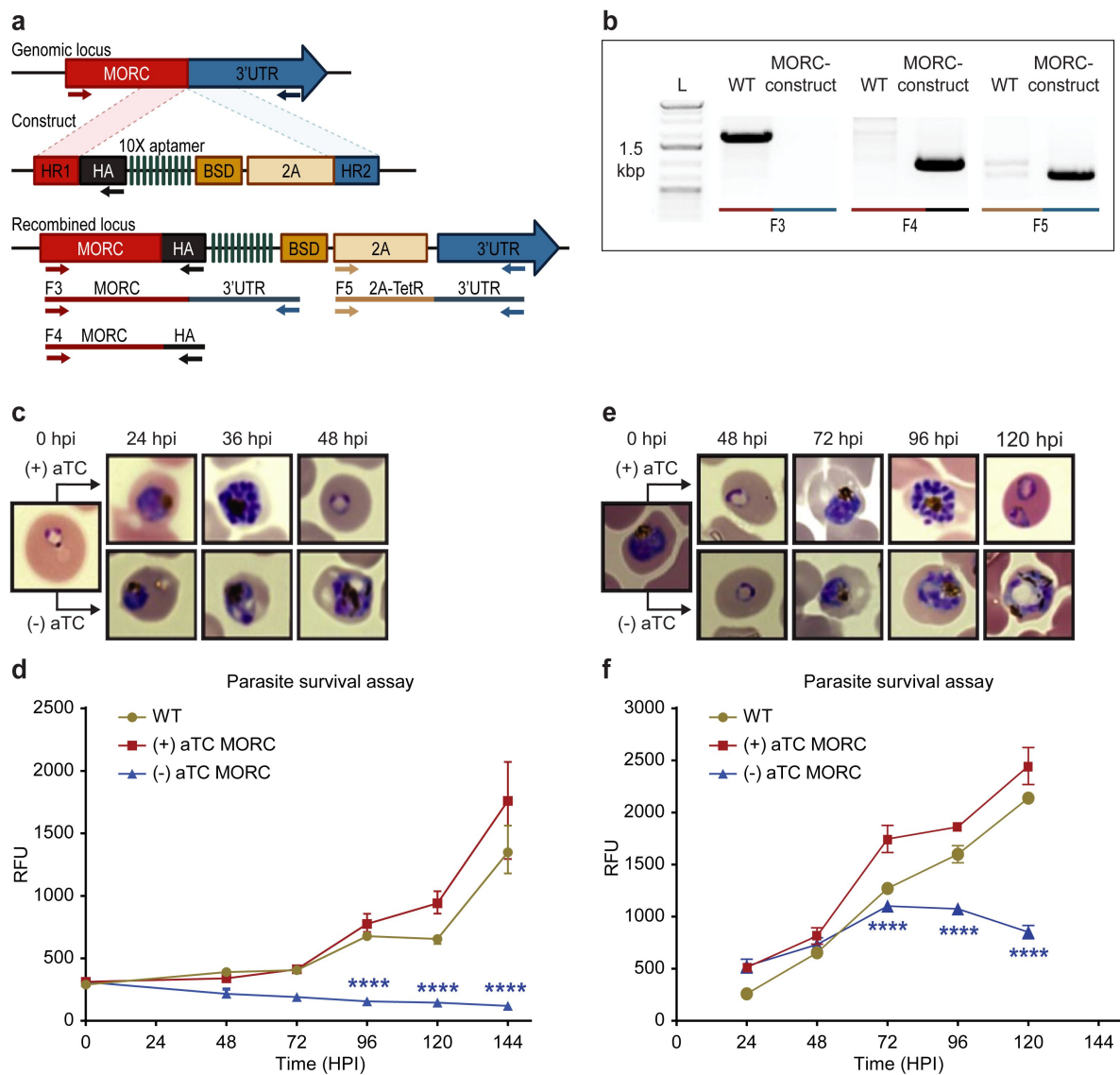


Fig. 3.

***Pfmorc* is essential for cell survival.**

(a) Diagram representation of *Pfmorc*-HA-TetR-DOZI plasmid. **(b)** PCR amplification is used to verify genomic insertion using primers sets targeting 1.5 kbp of WT *Pfmorc* genome locus absent in transgenic line (MORC construct) (F3) as well as verification of HA insertion (F4) and TetR-DOZI system extending along 3' UTR of the construct (F5). **(c)** Phenotypic and **(d)** quantitative analysis of parasite cell progression after aTC withdrawal at the ring stage (0-6 hpi) (2-way ANOVA, $n=3$, $p \leq 0.0001$). **(e)** Phenotypic and quantitative **(f)** analysis of parasite cell progression after aTC removal at the trophozoite stage of cell cycle progression (24 hpi) (2-way ANOVA, $n=3$, $p \leq 0.0001$).

However, clear indications of stress and cell cycle arrest were ultimately detected at trophozoite and early schizont stages of the second cell cycle. Quantitative analysis of (-) aTC *PfMORC* cultures revealed significantly decreased parasitemia of ~37% and 66% ($n=3$, $p \leq 0.0001$) at 96 hpi and 120 hpi, respectively, compared to (+) aTC *PfMORC* and WT control lines (**Fig. 3e-f**, Sup. data 5), confirming the importance of the protein at the trophozoite and early schizont stages.

Effect of *PfMORC* knockdown on parasite transcriptome

To define the effects of *PfMORC* KD on transcription, we conducted detailed time-course measurements of mRNA levels using RNA-sequencing (RNA-seq) throughout the parasite asexual cycle on (+/-) aTC *PfMORC* lines. Parasites were first synchronized and aTC was removed from one set of samples. Total RNA was then extracted at the trophozoite (24 hpi) and the schizont (36 hpi) stages to allow for detection of early and late changes in gene expression after aTC removal. Pairwise correlations analysis (**Fig. S4a**) between (+/-) aTC *PfMORC* treated lines at the two different time points confirmed appropriate reproducibility of our RNA-seq experiments.

Our results identified a relatively low number of differentially expressed genes at 24 hpi with 96 and 93 upregulated and downregulated genes, respectively ($FDR < 0.05$ and $\log_2 FC > 0.5$ or < -0.5) (**Fig 4a**, Sup. data 6). This data is in agreement with the absence of major phenotypic changes observed in (-) aTC *PfMORC* parasite cultures after 24 hours. GO enrichment analysis indicated upregulation of genes in response to xenobiotic stimulus including several phosphatases, hydrolases and heat shock proteins suggesting stress sensing of the parasites at this stage (**Fig. 4c**). GO analyses of downregulated genes, on the other hand, were found to be closely associated with regular metabolic processes and intracellular transport mechanisms suggesting dysregulation of regular cellular activity. Although, at this stage it is probable that the limited depletion of *PfMORC* (58%) has restricted effects at the transcriptional level (See **Fig. S3**) and that the observed down-regulated genes are likely due to an indirect effect of cell cycle arrest. At 36 hpi the number of genes that exhibit changes in gene expression were significantly higher with 1319 upregulated and 1150 downregulated ($FDR < 0.05$ and $\log_2 FC > 0.5$ or < -0.5) in (-) aTC *PfMORC* conditions, reflecting of a significant decrease of the protein (~ 88.2%) and confirming the significant changes observed at the phenotypic level (**Fig. 4b**). These results emphasize the cascading effects of *PfMORC* KD as the parasite progressed throughout its cell cycle. GO analysis indicated an enrichment of upregulated genes involved in multiple pathways including ATP metabolic process, mitochondria, translation, food vacuole and protein folding (**Fig. 4d**); characteristic of not only stress, but also a clear signal of cell cycle arrest in (-) aTC *PfMORC* samples. Among upregulated genes, we also observed several *var* genes and genes exported at the surface of the red blood cell that could be linked to significant decreased in *PfMORC* binding of these genes families as well as a disorganization of the heterochromatin cluster(s) at the trophozoite and schizont stages. GO enrichment analysis for genes that were downregulated included many genes involved in DNA replication, chromosome organization and mitotic spindle further emphasizing a strong cell cycle arrest and absence of potential major compensatory mechanisms for cell division and chromatin organization (**Fig. 4d**). Additional downregulated genes were required for invasion such as several merozoite surface proteins. Our results clearly indicate stress sensing and an arrest in cell cycle progression at the trophozoite and schizont stages (Sup. data 5)(59).

Although these results are compelling, the small overlap observed between the ChIP-seq signals and RNA-seq results indicate that major changes observed in gene expression at the schizont stages are not simply the result of reduced *PfMORC* binding in targeted gene bodies but a combination of direct and indirect effects of the degradation of *PfMORC* that leads to cell cycle arrest and potential collapse of the heterochromatin (**Fig. 4e**).

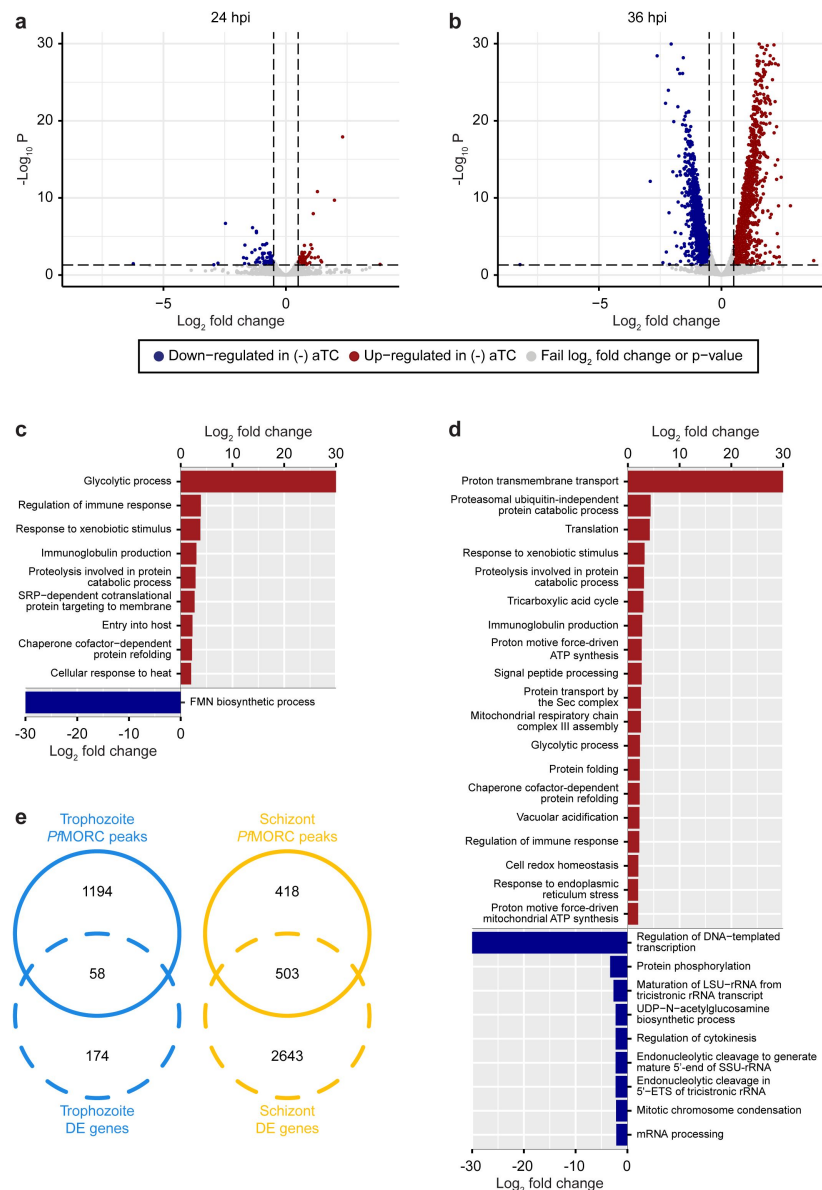


Fig. 4.

PfMORC KD on parasite transcriptome.

Volcano plots denoting upregulated (red), and downregulated (blue) genes discovered through differential expression analysis following *PfMORC* knockdown at (a) 24 hpi (b) 36 hpi. Gene ontology enrichment analysis for upregulated (red) and downregulated (blue) genes at (c) 24 hpi and (d) 36 hpi. (e) Overlap of differentially expressed genes and genes containing significant peaks called by *PfMORC* ChIP-seq analysis at the trophozoite and schizont stages.

PfMORC knockdown erodes antigenic-gene silencing framework

We next sought to analyze the effect of *PfMORC* down-regulation on the global chromatin landscape. We therefore performed a ChIP-seq experiment against histone H3K9me3 and H3K9ac marks in response to *PfMORC* depletion. Synchronized parasites were equally split between permissive and repressive conditions. At trophozoite (24 hpi) and schizont (36 hpi) stages of development, both control (+aTC) and experiment (-aTC) samples were fixed and parasites collected for chromatin immunoprecipitation followed by sequencing. The experimental procedure was performed in duplicate between (+/-) aTC *PfMORC* treated lines. Correlation analysis confirms the reproducibility of our ChIP-seq experiments.

Results revealed no significant changes in histone H3K9ac marks across the genome (data not shown) but a reduction in the heterochromatin landscape in the *PfMORC* depleted conditions, specifically in the telomere regions of the chromosomes (**Fig. 5a**). *PfMORC* KD resulted in significantly reduced (Mann-Whitney *U* test, $p < 0.05$) H3K9me3 marks within *var* gene promoters from 200-800 bp upstream of the TSS, as well as 400-800 bp downstream of the TSS (**Fig. 5b-c**). This result coincides with our transcriptomic profile and the upregulation of *var* genes at 36 hpi of (-)aTC samples (**Fig 4d**, Sup. data 6). Further analyses revealed similar trends in other gene families associated with parasite reinvasion and immune response typically silenced under WT conditions. Most noticeable was the reduced H3K9me3 coverage across the *rifin* gene family, which displayed a significant reduction ($p < 0.05$) in all bins from 1000 bp upstream of the TSS to 1000 bp downstream from the end of the gene (**Fig. 5b-c**) in response to *PfMORC* depletion. There is a clear reduction of H3K9me3 coverage of virulence genes shown to be upregulated following *PfMORC* knockdown (**Fig. 5c**). Overall, these findings confirm that *PfMORC* protein depletion not only affects chromatin landscape but has significant impact on heterochromatin. *PfMORC* downregulation leads to the erosion of heterochromatin integrity most likely required for mutually exclusive gene expression and immune evasion within heterochromatin clusters.

Knockdown of *PfMORC* expression results in the loss of tightly regulated heterochromatin structures

To better understand the effect that downregulation of *PfMORC* has on the chromatin structure and investigate whether changes in chromatin accessibility may explain large changes in gene expression observed using RNA-seq, we performed Hi-C on (+/-) aTC *PfMORC* cultures at 24 hpi (trophozoite) and 36 hpi (schizont). Biological replicates for each sample were collected and used to generate Hi-C libraries with >37 million reads per replicate/sample, ensuring comprehensive coverage of both intrachromosomal and interchromosomal interactions. After processing (pairing, mapping, and quality filtering) the raw sequences via HiC-Pro (60), there were approximately 15 million ($\sigma = \sim 10$ million) high quality interaction pairs per sample. Due to the high number of reads and relatively small size of the *P. falciparum* genome (23.3 Mb) compared to higher eukaryotes, we elected to bin our Hi-C data at 10-kb resolution, which allowed the identification of genome-wide patterns while not introducing much noise by binning at too high of a resolution. A high stratum-adjusted correlation (Fig. S4), especially at the 24 hpi time point, suggested that the chromatin structure was consistent between biological replicates, therefore we chose to combine replicates for downstream analyses and visualization.

Because of variation in the number of valid interaction pairs between (+/-) aTC *PfMORC* samples, 100 iterations of random sampling were performed on samples with higher read count to obtain ~35 million and ~9 million interaction pairs at 24 hpi and 36 hpi, respectively. ICED normalized intrachromosomal heatmaps displayed a high proportion of interactions at distances less than 10% the length of each chromosome as well as strong subtelomeric interactions and internal regions containing genes involved in antigenic variation (**Fig. 6a-b**, Fig. S6-S9) This pattern was similar to those observed previously at various stages of the life cycle (27, 60, 61) and

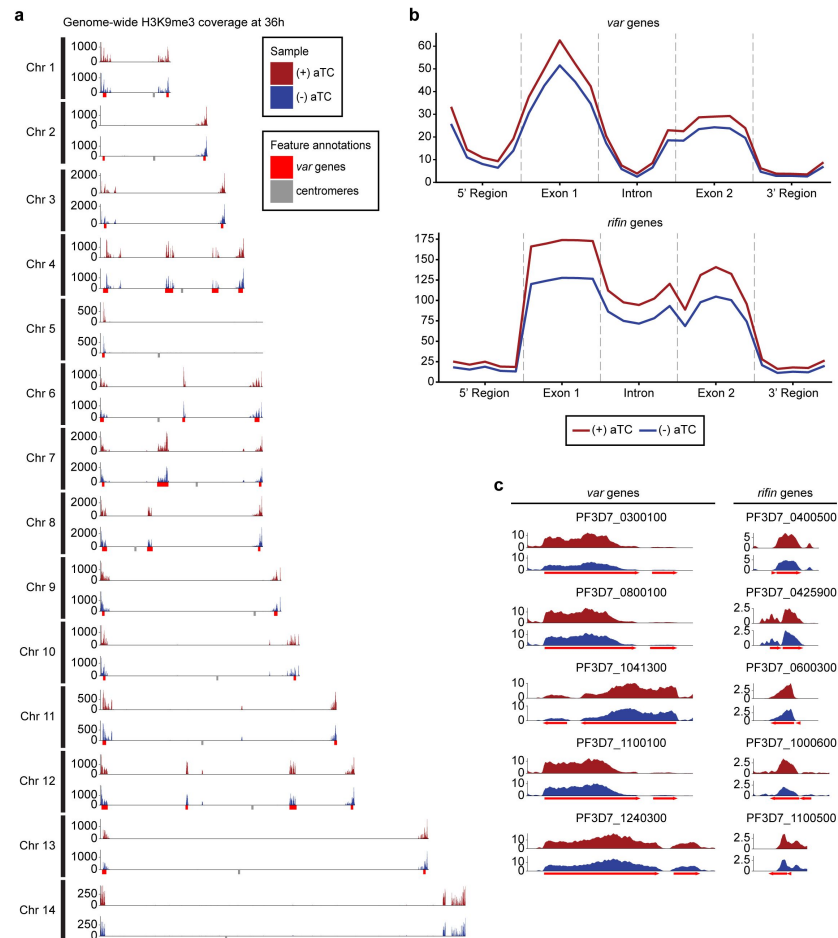


Fig. 5.

Impact of *PfMORC* KD on heterochromatin markers.

(a) Genome-wide H3K9me3 coverage of IGG subtracted and per-million normalized (+/-) aTC show similar distribution and concentration within telomeres and antigenic gene clusters highlighted in red. Replicates (n = 2) are merged using the mean of the normalized read coverage per base pair. **(b)** Binned coverage of *var* and *rifin* genes from 1 kb upstream of the TSS to 1 kb downstream of the end. The exons and intron of all genes within these families are split into five equal sized bins and the 5' and 3' regions are binned into five 200 bp bins. Read counts within each bin are per-million and bin length normalized prior to plotting. **(c)** Coverage of the five most upregulated *var* and *rifin* genes as determined by the transcriptomic analysis which shows elevated coverage in (+) aTC cells.

confirm that genes involved in antigenic variation are usually confined within dense heterochromatin rich regions to aid the tight control of *var* genes necessary for their mutually exclusive expression and immune evasion(53, 54). Data generated at 36 hpi in (-) aTC *PfMORC* parasite cultures were however tumultuous, with a decrease in defined heterochromatin borders and fewer long-range intrachromosomal interactions across the genome. This may indicate a significant loss of chromatin maintenance in (-) aTC *PfMORC* parasites (Fig. 6a-b).

To pinpoint which regions were most strongly affected by knockdown of *PfMORC*, we used Selfish(62) to identify differential intrachromosomal and interchromosomal interactions in the (+/-) aTC *PfMORC* samples. Although the 24 hpi and 36 hpi time points shared similarities in loci most highly affected by *PfMORC* KD on many chromosomes, the changes at 36 hpi were more significant with a higher log₂ FC (FDR < 0.05) at most loci (Fig. 6c-d, Fig. S10-S11). We also observed consistent loss of interactions between most regions containing *var* genes. These results indicate a failed attempt of the (-) aTC *PfMORC* parasites to maintain their overall chromatin structure with a significant weakening of the tightly controlled heterochromatin cluster.

We further validated our results using PASTIS to generate coordinate matrices and subsequently visualize a consensus three-dimensional model of the chromosome folding and their overall organization within the nucleus (63). This can indicate changes in spatial distance that best describe the interaction data. The overall change in the chromatin 3D structure was clearly shown by our models of (+/-) aTC *PfMORC* samples (Fig. 6e). While the co-localization of centromeres and telomeres in different regions of the nucleus were conserved, the 3D chromatin structure of (-) aTC *PfMORC* at 36 hpi displayed clear opening of the chromatin and loss of interactions in most regions. This could explain the large changes in gene expression, including increased expression of all *var* and *rifin* genes in the (-) aTC *PfMORC* line (Sup. data 6). While heterochromatin maintenance may be essential for tight control of *var* gene expression, preservation of the overall structure of the chromatin may be necessary to regulate the accurate expression of most *P. falciparum* transcripts.

Discussion

Despite intensive investigations on the dynamic nature of the *Plasmodium* chromatin, progress has been slow in capturing the regulatory factors that maintain and control chromatin structure and gene expression throughout parasite development. Preliminary studies performed by our lab and others have identified *PfMORC* as one of the most abundant chromatin-bound proteins(32, 35). MORC proteins from plant to animal cells have a broad range of binding sites within the genome and participate in either DNA methylation establishment or heterochromatin formation (30, 64–69). They have also been shown to co-localize with TFs in unmethylated promoter regions to alter chromatin accessibility and regulate TF binding and gene expression (66–69). For instance, AP2-P (PF3D7_1107800), AP2-G5, ISW1, NUP2, MCM5 and HDAC1 have been shown to interact with MORC in several published works (33, 42, 43), including this work. Protein pull-down experiments in *Plasmodium* spp. and *T. gondii* studies have detected MORC in complexes with several AP2 TFs. In *T. gondii*, *TgMORC* was characterized as an upstream transcriptional repressor of sexual commitment (33, 35). In *Plasmodium*, the role of *PfMORC* was still poorly understood. Conflicting results on the essentiality of *PfMORC* using either KD or KO strategies remained unresolved(32, 56). Using the CRISPR/Cas9 genome editing tool, we have determined the nuclear localization, genome-wide distribution, and regulatory impacts of *PfMORC* on the parasite chromatin, transcriptome, and cell cycle development. We now have powerful evidence demonstrating that *PfMORC* is not only critical for parasite cell cycle progression and its survival but has a direct role in heterochromatin formation and gene silencing, including regulating immune evasion through antigenic variation. Disparities observed in the essentiality of *PfMORC* for the parasite survival in previous studies(32) are most likely the results of weak protein disruption highlighting the need for a significant downregulation of *PfMORC* for true

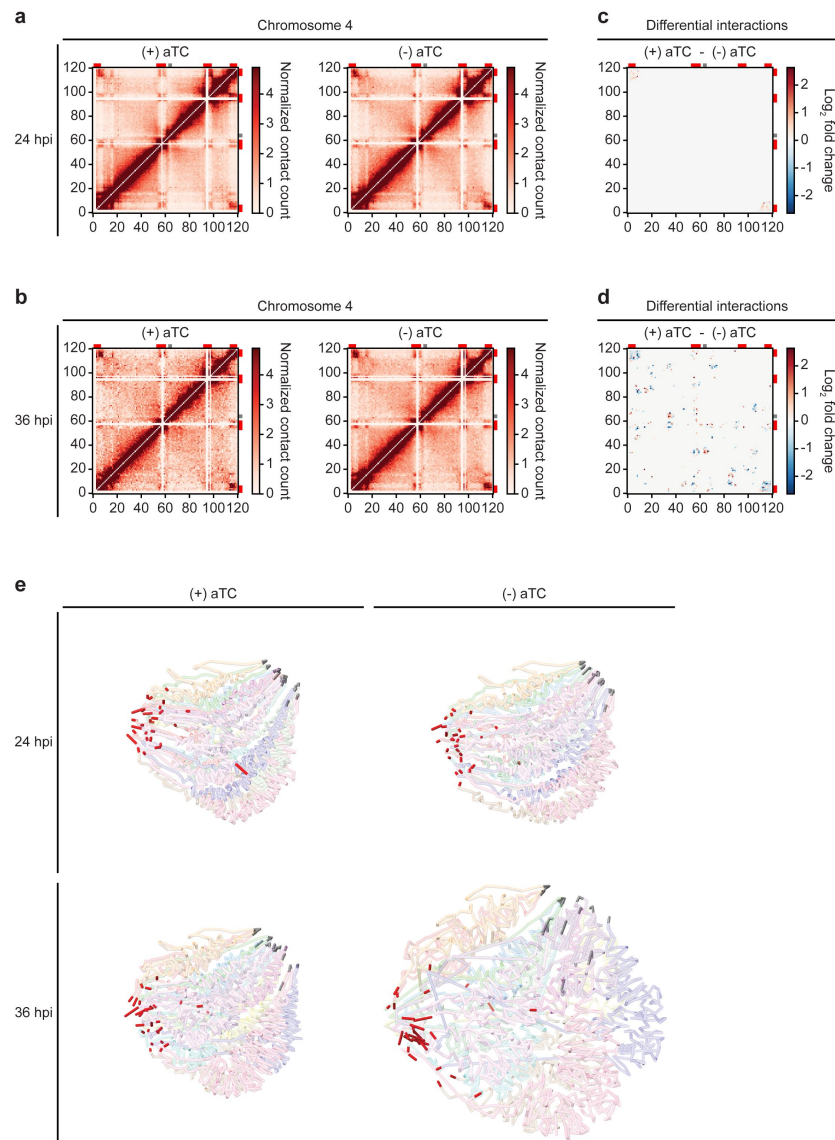


Fig. 6.

Loss of *Pf*MORC expression correlates with heterochromatin expansion.

Intrachromosomal interaction heatmaps of (+/-) aTC *Pf*MORC for chromosome 4 at **(a)** 24 hpi and **(b)** 36 hpi displaying heterochromatin clustering within antigenic (*var*, *rifin* and *stevor*) gene-dense regions (red). Differential interaction heatmaps highlight changes in chromatin structure following removal of aTC and subsequent *Pf*MORC knockdown at **(c)** 24 hpi and **(d)** 36 hpi. **(e)** Whole-genome 3D models of the chromatin structure at both time points (24 hpi and 36 hpi) and (+/-) aTC.

functional analysis. Using IFA and protein pull-downs, we have confirmed that *Pf*MORC localizes to the nucleus and interacts with multiple ApiAP2 TFs, chromatin remodelers, and epigenetic players associated with heterochromatin regions including H3K9me3, SIP2, HDAC1 and the ISWI chromatin-remodeling complex (SWIB/MDM2) (52) (Fig. 1). Although some discrepancies have been observed with other studies, mainly due to the fact that MORC protein was not used as the bait protein, our study validated the interaction of MORC with these partners, supporting that *Pf*MORC is in complex with key heterochromatin regulators. Using a combination of ChIP-seq, protein knock down, RNA-seq and Hi-C experiments, we also demonstrated that the MORC protein is essential for the tight regulation of gene expression. We can speculate that lack of MORC impacts heterochromatin and chromatin compaction, preventing access to gene promoters from TFs and the general transcriptional machinery in a stage specific manner. Although additional experiments will be required, our hypothesis is reinforced by the fact that downregulation of the *Pf*MORC significantly reduced H3K9me3 coverage in the heterochromatin cluster(s) and 5' flanking regions of *var* and *rifin* genes. While we were unable to confirm a direct role of *Pf*MORC in sexual conversion due to a nonsense mutation in AP2-G gene in our transfected lines, its strong interaction with AP2-G during the asexual cell cycle indicates that *Pf*MORC in combination with other epigenetic factors may most likely control AP2-G expression and sexual differentiation. It is also important to recognize that the many pathways affected at the transcriptional level throughout the asexual stages in (-) aTC *Pf*MORC lines are not only the direct results of *Pf*MORC downregulation and decrease of its targeted DNA binding site. They are most likely a combination of direct and indirect effects of *Pf*MORC KD warranted by the cell cycle arrest observed at the phenotypic level and the collapse of the chromatin organization confirmed using chromatin conformation capture experiment. These direct and indirect effects should be carefully considered, and a combination of functional genomic studies should be completed when interpreting changes in gene expression in mutant generated lines.

All together our work demonstrated that in addition to its direct role in heterochromatin formation and antigenic variation, *Pf*MORC may act as a potential repressor to control a set of parasite specific genes including genes involved in parasite egress and invasion, and antigenic variation between the trophozoite to schizont stage transition (52). As our data confirm the importance of *Pf*MORC and the parasite specificity of several interacting partners, it is tempting to speculate that drugs targeting these protein complexes could lead to novel antiparasitic strategies.

Methods

Asexual parasite Culture and maintenance

Asexual *P. falciparum* strain NF54 or 3D7 parasites (MRA-1000, MRA-102 respectively) were propagated in 5% of human O⁺ erythrocytes and 10 mL of RPMI-1640 medium containing 0.5% Albumax II (Invitrogen), 2 mM L-glutamine, 50 mg/L hypoxanthine, 25 mM HEPES, 0.225% NaHCO₃ and 10 mg/mL gentamicin. They were maintained at 37°C and gassed with a sterile mixture of 5% O₂, 5% CO₂ and 90% N₂ (70, 71). Parasite synchronization was achieved using two 5% D-sorbitol treatments 8 hours apart (70, 71).

Plasmid Construction

Flagging of the *P. falciparum* MORC (Pf3D7_1468100) gene spanning position (Ch14: 2785386 - 2797934 (-)) was performed using a two-plasmid design to insert a 3x HA tag. The pCasG-Cas9-sgRNA plasmid vector (gifted by Dr. Sean Prigge) contains the site to express the sgRNA, along with the yDHODH gene as the positive selection marker. The gRNA oligos (Sup. data 1) were ligated after digestion of the plasmid with BsaI. The pDC2-cam-Cas9-U6 plasmid (gifted by Dr. Marcus Lee) was digested with BamHI and Apal to remove the eGFP tag from the backbone. 453 bp of C-terminal region of *Pf*MORC and 469 bp of 3'UTR were amplified from *P. falciparum* genomic DNA with their respective primers (Sup. data 1). A 3x HA-tag was fused to the C-terminal region of the

amplified product along with the 3'UTR formed through Gibson assembly master mix (NEB, E2611S). To generate the *Pf*MORC-HA knockdown constructs, a pKD^{*Pf*AUBL} plasmid (gifted by Dr. Sean Prigge)(72 [DOI](#)) was digested with AscI and AatII. Homology arms HA1 and HA2 of *Pf*MORC were amplified with respective primers (Sup. data 1) having 20 bp overhang and inserted into the digested plasmid using Gibson assembly mix. The resulting pKD-MORC-HA-TetR-DOZI construct was linearized with EcoRV enzyme prior to transfection. All constructs were confirmed through restriction enzyme digestion and Sanger sequencing.

Plasmids were isolated from 250 mL cultures of *Escherichia coli* (XL10-Gold Ultracompetent Cells, Agilent Cat. 200314) and 60 µg of pDC2-MORC-HA or linearized pKD-MORC-HA-TetR-DOZI, were used with 60 µg of pCasG-plasmid containing gRNA to transfect 200 µl of fresh red blood cells (RBCs) infected with 3-5% early ring stage parasites. After one erythrocytic cycle, transfected cultures were supplemented with 1.5 µM WR99210 (2.6nM) (provided by the (Jacobus Pharmaceuticals, Princeton, NJ) and 2.5 µg/mL Blasticidin (RPI Corp B12150-0.1). *Pf*MORC-HA-TetR-DOZI transfected parasites were maintained with 500 nM anhydrotetracycline (aTC)(45 [DOI](#), 58 [DOI](#)). Media and drug selection was replenished every 24 hours for 7 consecutive days after which DSM-1 drug selection was halted Once parasites were detected by microscopy, integration of the insert was confirmed by PCR amplification. To generate genetically homogenous parasite lines, the transfected parasites were serially diluted to approximately 0.5% parasite/well, into 96 well plates.

Molecular analysis of the transgenic lines

Genomic DNA (gDNA) was extracted and purified using the DNeasy Blood & Tissue kit (Qiagen) following instructions from the manufacturer. The diagnostic PCR analysis was used to genotype the transfected lines using the primers listed in Supplementary data 1. The PCR amplification was conducted using KAPA HiFi HotStart ReadyMix (Roche) and amplicons were analyzed by gel electrophoresis followed by sequencing. For whole genome sequencing, genomic DNA was fragmented using a Covaris S220 ultrasonicator and libraries were generated using KAPA LTP Library Preparation Kit (Roche, KK8230). To verify that the insertion was present in the genome at the correct location in both transfected lines, reads were mapped using Bowtie2 (v2.4.4) to the *P. falciparum* 3D7 reference genome (PlasmoDB, v48), edited to include the insertion sequence in the intended location. Integrative Genomic Viewer (IGV, Broad Institute) was used to verify that reads aligned to the modified sequence.

Variant analysis by genome-wide sequencing

To call variants (SNPs/indels) in the transfected lines compared to a previously sequenced control 3D7 line, genomic DNA reads were first trimmed to adapters and aligned to the *Homo sapiens* genome (assembly GRCh38) to remove human-mapped reads. Remaining reads were aligned to the *P. falciparum* 3D7 genome using bwa (version 0.7.17) and PCR duplicates were removed using PicardTools (Broad Institute). GATK HaplotypeCaller (<https://gatk.broadinstitute.org/hc/en-us> [DOI](#)) was used to call variants between the sample and the 3D7 reference genome for both the transfected lines and the NF54 control. Only variants that were present in both transfected lines but not the NF54 control line were kept. We examined only coding-region variants and removed those that were synonymous variants or were located in *var*, *rifin*, or *stevor* genes. Quality control of variants was done by hard filtering using GATK guidelines.

Immunofluorescence assays

A double-staining immunofluorescence assay was used on NF54 control and transgenic parasite lines of mixed parasite population growth stages. Parasites were washed in incomplete medium prior to fixing onto coverslips with 4% paraformaldehyde for 20 min at RT under darkness. After fixation, samples were washed 3-5 times with 1x PBS followed by permeabilization with 0.5%Triton X-100 in PBS for 25 min at RT. Subsequently, samples were subjected to PBS washes and then incubated overnight at 4°C in blocking buffer (2mg/ml BSA solution in PBS containing 0.05%

Tween-20 solution). Following overnight blocking, samples were washed and incubated for 1 hr at RT with Anti-HA Rb Ab (1:500, Abcam, ab9110) in blocking buffer. After primary Ab incubation, samples were subject to 3x washes with wash buffer (1x PBS containing 0.05% Tween-20) followed by incubation with anti-rabbit DyLight 550 (Abcam ab98489; 1: 500) secondary antibody for 1 h at room temperature. After incubation and series of washes with wash buffer, slides were incubated with anti-H3K9me3 antibody, Alexa Fluor 488 conjugate (Millipore 07-442-AF488; 1:100) for 1 hr at RT. Slides are then washed and mounted in Vectashield Antifade Mounting Medium with DAPI (Vector Laboratories, H-1200). Images were acquired using a Keyence BZ-X810 Fluorescence Microscope and were processed through ImageJ.

Western blotting

Parasites were synchronized twice at 8 hr. intervals between synchronizations. After one cycle, culture was washed 3x with complete media to remove residual aTC followed by dividing the cultures into two equal conditions. Parasites were grown with or without 500 nM aTC for 24 and 36 hrs. after which RBCs were lysed using 0.15 % saponin and parasites were collected after being washed with ice cold 1X PBS. Proteins were recovered from the lysed parasites after 30 min of incubation in lysis buffer (150 mM NaCl, 0.5 % NP40, 50 mM Tris pH 8, 1 mM EDTA and protease inhibitors) and 10 sec of sonication. Proteins were quantified with Pierce™ BCA Protein Assay Kit (Thermo Fisher 23227). 20 µg of proteins were loaded onto 3–8% Criterion XT Tris-Acetate Midi Protein Gels (Bio-Rad, 3450129). After migration, proteins were transferred onto a PVDF membrane and the membrane was blocked and then probed overnight with anti-HA tag antibody (1:2500, Abcam, ab9110) as well as anti-aldolase antibody as a loading control (1:10000, abcam, ab252953). After primary Ab incubation, blots were subsequently washed 3x with a washing buffer followed by HRP-labeled Goat anti-Rabbit IgG (H + L) (1:10,000, Novex™, A16104). Clarity™ Western ECL Substrate (Bio-Rad, 1705060) was applied to reveal the blots. Relative abundance of (+/-) aTC *Pf*MORC was calculated by Bio-Rad ChemiDoc imagelab software.

Immunoprecipitation followed by MudPIT mass spectrometry

Mid-to late-stage asexual parasites were collected following saponin treatment and purified samples were then resuspended into fresh IP buffer (50 mM Tris-HCl pH 7.5, 300 mM NaCl, 0.5 mM EDTA, 0.5 mM EGTA, 2 mM AEBSF 0.5 % Triton X-100, and EDTA-free protease inhibitor cocktail (Roche)). Post cell lysis solution was homogenized via sonication for 6-9 rounds. The soluble extracts were centrifuged at 13,000 x g for 15 min at 4°C. The lysates were precleared with Dynabeads™ Protein A (Invitrogen) for 1h at 4°C. Anti-HA tag antibody (1:2500, Abcam, ab9110) are added to control and HA tagged *Pf*MORC precleared protein extract samples for 1hr at 4°C under constant rotation followed by the addition of fresh Dynabeads™ Protein A beads to each sample and incubated overnight at 4°C. Dynabeads were washed 3 times with 500 µL of buffer (PBS, 0.05% Tween-20). Proteins were eluted into an elution buffer (50 mM Tris-HCl pH 6.7, 100 mM DTT and 2% SDS). The eluent was subsequently precipitated overnight in 20% TCA followed by cold acetone washes. The urea-denatured, reduced, alkylated, and digested proteins were analyzed by Multidimensional Protein Identification Technology (MudPIT) on a Orbitrap Elite mass spectrometer coupled to an Agilent 1260 series HPLC, as described previously (73 [↗](#)).

Proteomics data processing and analysis

Tandem mass (MS/MS) spectra were interpreted using ProluCID v.1.3.3 (74 [↗](#)) against a database consisting of 5527 non-redundant (NR) *Plasmodium falciparum* 3D7 proteins (PlasmoDB, v42), 36661 NR human proteins (NCBI, 2018-03-30 release), 419 common contaminants (human keratins, IgGs, and proteolytic enzymes), together with shuffled versions of all of these sequences. DTASelect v.1.9 (75 [↗](#)) and swallow v.0.0.1, an in-house developed software (<https://github.com/tzw-wen/kite> [↗](#)) were used to control FDRs resulting in protein FDRs less than 1.86%. All datasets were contrasted against their merged data set, respectively, using Contrast v1.9 (75 [↗](#)) and in-house developed sandmartin v.0.0.1 (<https://github.com/tzw-wen/kite/tree/master/> [↗](#) kitelinux). Our in-

house developed software, NSAF7 v.0.0.1 (<https://github.com/tzw-wen/kite/tree/master/windowsapp/NSAF7x64>), was used to generate spectral count-based label free quantitation results (76). QPROT(49, 77) was used to calculate values of $\log_2$ fold change and Z-statistic to compare two replicate *Pf*MORC affinity purifications to two negative controls. Proteins enriched in the *Pf*MORC-HA samples with values of $\log_2$ fold change ≥ 2 and Z-statistic ≥ 5 were considered significantly enriched.

ChIP assay

*Pf*MORC-HA and *Pf*MORC KD asexual stage parasites were supplemented with and without aTC along with NF54 parasites (as a control) and were harvested at 24 and 36 HPS and cross linked with 1% formaldehyde for 10 min at 37°C followed by quenching with 150 mM glycine and 3 x washing with 1 x PBS. The pellets were resuspended in 1 mL of nuclear extraction buffer (10 mM HEPES, 10 mM KCl, 0.1 mM EDTA, 0.1 mM EGTA, 1 mM DTT, 0.5 mM AEBSF and 1x Protease inhibitor cocktail), and incubated on ice for 30 min before addition of NP-40/Igepal to a final of 0.25%. After lysis, the parasites are subject to homogenization by passing through a 26G 3/8 needle/syringe to burst the nucleus. After centrifugation for 20 min at 2,500 x g at 4°C, samples were resuspended in 130 μ L of shearing buffer (0.1% SDS, 1 mM EDTA, 10 mM Tris-HCl pH 7.5 and 1X Protease inhibitor cocktail) and transferred to a 130 μ L Covaris sonication microtube. The samples were then sonicated using a Covaris S220 Ultrasonicator for 8 min (Duty cycle: 5%, intensity peak power:140, cycles per burst: 200, bath temperature: 6°C). The samples were transferred to ChIP dilution buffer (30 mM Tris-HCl pH 8.0, 3 mM EDTA, 0.1% SDS, 30 mM NaCl, 1.8% Triton X-100, 1X protease inhibitor tablet, 1X phosphatase inhibitor tablet) and centrifuged for 10 min at 13,000 rpm at 4°C, retaining the supernatant. For each sample, 13 μ L of protein A agarose/salmon sperm DNA beads were washed three times with 500 μ L ChIP dilution buffer by centrifuging for 1 min at 1,000 rpm at room temperature. For pre-clearing, the diluted chromatin samples were added to the beads and incubated for 1 hour at 4°C with rotation, then pelleted by centrifugation for 1 min at 1,000 rpm. Before adding antibody, ~10% of each sample was taken as input. 2 μ g of anti-HA tag (1:2500, Abcam, ab9110) or rabbit polyclonal anti-H3K9me3 or anti-H3K9ac (Millipore no. 07–442 and H0913) antibodies were added to the sample and incubated overnight at 4°C with rotation. For each sample, rabbit IgG antibody (Cat.689 No. 10500c, Invitrogen) was used as a negative-control library. Per sample, 25 μ L of protein A agarose/salmon sperm DNA beads were washed with ChIP dilution buffer (no inhibitors), blocked with 1 mg/mL BSA for 1 hour at 4°C, and then washed three more times with buffer. 25 μ L of washed and blocked beads were added to the sample and incubated for 1 hour at 4°C with continuous mixing to collect the antibody/protein complex. Beads were pelleted by centrifugation for 1 min at 1,000 rpm at 4°C. The bead/antibody/protein complex was then washed with rotation using 1 mL of each buffer twice; low salt immune complex wash buffer (1% SDS, 1% Triton X-100, 2 mM EDTA, 20 mM Tris-HCl pH 8.0, 150 mM NaCl), high salt immune complex wash buffer (1% SDS, 1% Triton X-100, 2 mM EDTA, 20 mM Tris-HCl pH 8.0, 500 mM NaCl), and TE wash buffer (10 mM Tris-HCl pH 8.0, 1 mM EDTA). Complexes were eluted from antibodies by adding 250 μ L of freshly prepared elution buffer (1% SDS, 0.1 M sodium bicarbonate). 5 M NaCl were added to the eluate and cross-linking was reversed by heating at 45°C overnight followed by addition of 15 μ L of 20 mg/mL RNase A with 30 min incubation at 37°C. After this, 10 μ L of 0.5 M EDTA, 20 μ L of 1 M Tris-HCl pH 7.5, and 2 μ L of 20 mg/mL proteinase K were added to the eluate and incubated for 2 hours at 45°C. DNA was recovered by phenol/chloroform extraction and ethanol precipitation. DNA was purified using Agencourt AMPure XP beads. Libraries were then prepared from this DNA using KAPA LTP Library Preparation Kit (Roche, KK8230) and sequenced on a NovaSeq 6000 machine.

ChIP-seq analysis

FastQC (version 0.11.8, <https://www.bioinformatics.babraham.ac.uk/projects/fastqc/>) was used to analyze raw read quality. Any adapter sequences were removed using Trimmomatic (version 0.39, <http://www.usadellab.org/cms/?page=trimmomatic>). Bases with Phred quality scores below 20 were trimmed using Sickle (version 1.33, <https://asexualstageparasitesb.com/najoshi/sickle>). The

resulting reads were mapped against the *P. falciparum* genome (v48) using Bowtie2 (version 2.4.4). Using Samtools (version 1.11), only properly paired reads with mapping quality 40 or higher were retained and reads marked as PCR duplicates were removed by PicardTools MarkDuplicates (version 2.18.0, Broad Institute). Genome-wide read counts per nucleotide were normalized by dividing millions of mapped reads for each sample (for all samples including input) and subtracting IgG read counts from the anti-HA IP counts. Peak calling was performed using MACS3 (version 3.0.0a7) (78) with the upper and lower limits for fold enrichment set to 2 and 50, respectively.

Phenotypic analyses

P. falciparum NF54 line along with transgenic *Pf*MORC-HA-TetR-DOZI line were synchronized, grown for one cycle post-synchronization and subject to 3 consecutive washes with 1X PBS before being supplemented either with or without (aTC) for 48 hrs. or 72 hrs. (depending on which stage the parasites were washed). Culture media were replaced daily with or without aTC supplementation.

Quantitative Growth assay

Synchronous cultures either ring or trophozoite stage at 0.5 % parasitemia were grown with or without aTC (500 nM) into a 96 well plate in triplicates for each time point. Parasites were collected at every 24 hrs. time for 3 cycles and subjected to SYBR green assay (Thermo Fisher, S7523).

RNA-seq library preparation

Parasites at ring, trophozoite or schizont stage were extracted following saponin treatment before flash freezing. Two independent biological replicates were generated for each time point, culture condition and line. Total RNA was extracted with TRIzol® LS Reagent (Invitrogen) followed by incubation for 1 hr with 4 units of DNase I (NEB) at 37°C. RNA samples were visualized by RNA electrophoresis and quantified on Synergy™ HT (BioTek). mRNA was then purified using NEBNext® Poly(A) mRNA Magnetic Isolation Module (NEB, E7490) according to the manufacturer's instructions. Libraries were prepared using NEBNext® Ultra™ Directional RNA Library Prep Kit (NEB, E7420) and amplified by PCR with KAPA HiFi HotStart Ready Mix (Roche). PCR conditions consisted of 15 min at 37°C followed by 12 cycles of 98°C for 30 sec, 55°C for 10 s and 62°C for 1 min 15 sec, then finally one cycle for 5 min at 62°C. The quantity and quality of the final libraries were assessed using a Bioanalyzer (Agilent Technology Inc). Libraries were sequenced using a NovaSeq 6000 DNA sequencer (Illumina), producing paired-end 100-bp reads.

RNA-seq data processing and differential expression analysis

FastQC [https://www.bioinformatics.babraham.ac.uk/projects/fastqc/79] was used to analyze raw read quality and thus 11 bp of each read and any adapter sequences were removed using Trimmomatic (v0.39) [http://www.usadellab.org/cms/?page=trimmomatic80]. Bases were trimmed from reads using Sickle with a Phred quality threshold of 25 (v1.33) [https://github.com/najoshi/sickle81] and reads shorter than 18 bp were removed. The resulting reads were mapped against the *P. falciparum* 3D7 genome (PlasmoDB, v53) using HISAT2 (v2-2.2.1) with default parameters. Uniquely mapped, properly paired reads with mapping quality of 40 or higher were retained using SAMtools (v1.11) [http://samtools.sourceforge.net/82]. Raw read counts were determined for each gene in the *P. falciparum* genome using BedTools [https://bedtools.readthedocs.io/en/latest/83] to intersect the aligned reads with the genome annotation. Differential expression analysis was performed using DESeq2 to call up- and down-regulated genes (FDR < 0.05 and log₂ FC > 0.5). Volcano plots were made using the R package EnhancedVolcano.

Hi-C library preparation

(+/-) aTC parasites at 24 hpi and 36 hpi were cross linked using 1.25% formaldehyde for 15 minutes at 37°C in 10 mL total volume followed by quenching with 150 mM glycine. Parasites were then washed three times with chilled 1x PBS on a rocking platform. Parasite nuclei were released using lysis buffer (10 mM Tris-HCl, pH 8.0, 10 mM NaCl, 2 mM AEBSF, 0.25% Igepal CA-630, and 1X EDTA-free protease inhibitor cocktail (Roche)) and 15 syringe passages through a 26.5-gauge needle. After releasing the crosslinked chromatin with 0.1% SDS, chromatin was digested using MboI restriction enzyme overnight at 37°C. Thereafter Hi-C libraries were prepared as previously described (27, 79).

Hi-C data processing, differential interaction analysis and generation of 3D models

Paired-end Hi-C libraries were processed (pairing, mapping, quality filtering, binning, and normalization) using HiC-Pro (60). A mapping quality cutoff of 30 was set while aligning to the *P. falciparum* genome (PlasmoDB, v58) and the resulting reads were binned at 10 kb resolution and ICED-normalized. Stratum-adjusted correlation coefficients were calculated using HiCRep (80), then replicates were merged to generate a single consensus sample for each time point and condition. The normalized matrices were per-million read count normalized and maximum values for the heatmap color scale on each chromosome was set to the maximum value for all samples to allow for direct comparisons between each condition and time point. Furthermore, due to the overrepresentation of short-range interactions, the maximum values for each heatmap were also set to the 90th percentile of each chromosome to slightly compress highly interacting regions and enhance visualization of interactions. Significant intrachromosomal interactions were identified with FitHiC (81) to identify the log-linear relationship between contact probability and genomic distance. Differential interchromosomal and intrachromosomal interactions between (+) aTC and (-) aTC were identified using Selfish (62) with default parameters (FDR < 0.05). The maximum and minimum values for the color scales in the differential heatmaps were set to the absolute value of the largest log₂ fold change within each chromosome. PASTIS (63) was used to generate coordinate matrices from the raw read count matrices output by HiC-Pro, and then visualized as 3D chromatin models in ChimeraX (82), highlighting regions containing *var* genes, telomeres, and centromeres.

Statistical analyses

Parasitemia and proportion of asexual stages were analyzed using a two-way ANOVA with Tukey's test for multiple comparisons. Significant differences were indicated as following: * for $p < 0.05$; ** for $p < 0.01$, *** for $p < 0.001$ and **** for $p < 0.0001$. Statistical tests were performed with GraphPad Prism. Figures were generated with GraphPad Prism and BioRender.

Data Availability

Sequence reads for all sequencing experiments have been deposited in the NCBI Sequence Read Archive with accession PRJNA994684. Original data underlying this manuscript generated at the Stowers Institute can be accessed from the Stowers Original Data Repository at <http://www.stowers.org/research/publications/LIBPB-2404>. The mass spectrometry dataset generated for this study is available from the MassIVE data repository (<ftp://MSV000092353@massive.ucsd.edu>) using the identifier MSV000092353. All other data are available in the main text, methods, and supplementary data.

Acknowledgements

This work was supported by NIH grants to KGLR (nos. 1R01 AI136511 and R21 AI142506-01) and by the University of California, Riverside to KGLR (no. NIFA-Hatch-225935). This publication includes data generated at the UC San Diego IGM Genomics Center utilizing an Illumina NovaSeq 6000 that was purchased with funding from a National Institutes of Health SIG grant (#S10 OD026929).

Author contributions

Conceptualization was the responsibility of KGLR who also supervised the project together with LF. MG generated all the constructs and cell lines used in the study. Immunofluorescence and growth assays were performed by MG and ZC. ChIP seq, RNA seq libraries were generated by MG and analyzed by TL and SA. Hi-C experiments were performed by MG and the data was analyzed by TL. IP-MS experiments were performed by MG, ZC, CB, and analyzed by TH, CB and LF. Software and data curation were provided by TL, SA, and CB. ZC, TH, TL and KGLR wrote the original draft. All authors reviewed and edited the final version of the manuscript.

Declarations of Interest

The authors declare no competing interests.

References

1. WHO (2023) **World malaria report 2023** WHO
2. Coulson R. M., Hall N., Ouzounis C. A. (2004) **Comparative genomics of transcriptional control in the human malaria parasite *Plasmodium falciparum*** *Genome Res* **14**:1548–1554
3. Balaji S., Babu M. M., Iyer L. M., Aravind L. (2005) **Discovery of the principal specific transcription factors of Apicomplexa and their implication for the evolution of the AP2-integrase DNA binding domains** *Nucleic Acids Res* **33**:3994–4006
4. De Silva E. K., et al. (2008) **Specific DNA-binding by apicomplexan AP2 transcription factors** *Proc Natl Acad Sci U S A* **105**:8393–8398
5. Yuda M., et al. (2009) **Identification of a transcription factor in the mosquito-invasive stage of malaria parasites** *Mol Microbiol* **71**:1402–1414
6. Iwanaga S., Kaneko I., Kato T., Yuda M. (2012) **Identification of an AP2-family protein that is critical for malaria liver stage development** *PLoS One* **7**
7. Sinha A., et al. (2014) **A cascade of DNA-binding proteins for sexual commitment and development in *Plasmodium*** *Nature* **507**:253–257
8. Kafsack B. F., et al. (2014) **A transcriptional switch underlies commitment to sexual development in malaria parasites** *Nature* **507**:248–252
9. Lesage K. M., et al. (2018) **Cooperative binding of ApiAP2 transcription factors is crucial for the expression of virulence genes in *Toxoplasma gondii*** *Nucleic Acids Res* **46**:6057–6068
10. Yuda M., Iwanaga S., Shigenobu S., Kato T., Kaneko I. (2010) **Transcription factor AP2-Sp and its target genes in malarial sporozoites** *Mol Microbiol* **75**:854–863
11. Modrzynska K., et al. (2017) **A Knockout Screen of ApiAP2 Genes Reveals Networks of Interacting Transcriptional Regulators Controlling the *Plasmodium* Life Cycle** *Cell Host Microbe* **21**:11–22
12. Gu C., et al. (2017) **Multiple regulatory roles of AP2/ERF transcription factor in angiosperm** *Bot Stud* **58**
13. Yuda M., Kaneko I., Iwanaga S., Murata Y., Kato T. (2020) **Female-specific gene regulation in malaria parasites by an AP2-family transcription factor** *Mol Microbiol* **113**:40–51
14. Yuda M., Iwanaga S., Kaneko I., Kato T. (2015) **Global transcriptional repression: An initial and essential step for *Plasmodium* sexual development** *Proc Natl Acad Sci U S A* **112**:12824–12829
15. Bozdech Z., et al. (2003) **Expression profiling of the schizont and trophozoite stages of *Plasmodium falciparum* with a long-oligonucleotide microarray** *Genome Biol* **4**

16. Le Roch K. G., et al. (2003) **Discovery of gene function by expression profiling of the malaria parasite life cycle** *Science* **301**:1503–1508
17. Bozdech Z., et al. (2003) **The transcriptome of the intraerythrocytic developmental cycle of *Plasmodium falciparum*** *PLoS Biol* **1**
18. Srinivasan P., et al. (2004) **Analysis of the *Plasmodium* and *Anopheles* transcriptomes during oocyst differentiation** *J Biol Chem* **279**:5581–5587
19. Silvestrini F., et al. (2005) **Genome-wide identification of genes upregulated at the onset of gametocytogenesis in *Plasmodium falciparum*** *Mol Biochem Parasitol* **143**:100–110
20. Yang M., et al. (2021) **Full-Length Transcriptome Analysis of *Plasmodium falciparum* by Single-Molecule Long-Read Sequencing** *Front Cell Infect Microbiol* **11**
21. Dekker J., Rippe K., Dekker M., Kleckner N. (2002) **Capturing chromosome conformation** *Science* **295**:1306–1311
22. Dostie J., et al. (2006) **Chromosome Conformation Capture Carbon Copy (5C): a massively parallel solution for mapping interactions between genomic elements** *Genome Res* **16**:1299–1309
23. Cui L., Fan Q., Cui L., Miao J. (2008) **Histone lysine methyltransferases and demethylases in *Plasmodium falciparum*** *Int J Parasitol* **38**:1083–1097
24. Dixon J. R., et al. (2012) **Topological domains in mammalian genomes identified by analysis of chromatin interactions** *Nature* **485**:376–380
25. Dembele L., et al. (2014) **Persistence and activation of malaria hypnozoites in long-term primary hepatocyte cultures** *Nat Med* **20**:307–312
26. Deng X., et al. (2015) **Bipartite structure of the inactive mouse X chromosome** *Genome Biol* **16**
27. Bunnik E. M., et al. (2019) **Comparative 3D genome organization in apicomplexan parasites** *Proc Natl Acad Sci U S A* **116**:3183–3192
28. Batugedara G., et al. (2020) **The chromatin bound proteome of the human malaria parasite** *Microb Genom* **6**
29. Lorkovic Z. J. (2012) **MORC proteins and epigenetic regulation** *Plant Signal Behav* **7**:1561–1565
30. Li D. Q., Nair S. S., Kumar R. (2013) **The MORC family: new epigenetic regulators of transcription and DNA damage response** *Epigenetics* **8**:685–693
31. Weiser N. E., et al. (2017) **MORC-1 Integrates Nuclear RNAi and Transgenerational Chromatin Architecture to Promote Germline Immortality** *Dev Cell* **41**:408–423
32. Singh S., et al. (2021) **The PfAP2-G2 transcription factor is a critical regulator of gametocyte maturation** *Mol Microbiol* **115**:1005–1024
33. Farhat D. C., et al. (2020) **A MORC-driven transcriptional switch controls *Toxoplasma* developmental trajectories and sexual commitment** *Nat Microbiol* **5**:570–583

34. Wang H., Zhang L., Luo Q., Liu J., Wang G. (2021) **MORC protein family-related signature within human disease and cancer** *Cell Death Dis* **12**
35. Zhong Z., et al. (2023) **MORC proteins regulate transcription factor binding by mediating chromatin compaction in active chromatin regions** *Genome Biol* **24**
36. Saksouk N., et al. (2005) **Histone-modifying complexes regulate gene expression pertinent to the differentiation of the protozoan parasite *Toxoplasma gondii*** *Mol Cell Biol* **25**:10301–10314
37. Bougdour A., et al. (2009) **Drug inhibition of HDAC3 and epigenetic control of differentiation in Apicomplexa parasites** *J Exp Med* **206**:953–966
38. Hillier C., et al. (2019) **Landscape of the Plasmodium Interactome Reveals Both Conserved and Species-Specific Functionality** *Cell Rep* **28**:1635–1647
39. Jeninga M. D., Quinn J. E., Petter M. (2019) **ApiAP2 Transcription Factors in Apicomplexan Parasites** *Pathogens* **8**
40. Hiyoshi T., Wada J. A. (1990) **Feline amygdaloid kindling and the sleep-waking pattern: observations on daily 22-hour polygraphic recording** *Epilepsia* **31**:131–138
41. Singh M. K., et al. (2021) **A nuclear protein, PfMORC confers melatonin dependent synchrony of the human malaria parasite *P. falciparum* in the asexual stage** *Sci Rep* **11**
42. Subudhi A. K., et al. (2023) **DNA-binding protein PfAP2-P regulates parasite pathogenesis during malaria parasite blood stages** *Nat Microbiol* **8**:2154–2169
43. Bryant J. M., et al. (2020) **Exploring the virulence gene interactome with CRISPR/dCas9 in the human malaria parasite** *Mol Syst Biol* **16**
44. Kumar Singh Maneesh, Tojal Da Silva Israel, Feijoli Santiago Verônica, Miriam S. Moraes, Jack Adderley, Christian Doerig, Giuseppe Palmisano, Manuel Llinás, Célia R. S Garcia (2023) **A Plasmodium falciparum MORC protein complex modulates epigenetic control of gene expression through interaction with heterochromatin** *eLife* **12**
45. Ganesan S. M., Falla A., Goldfless S. J., Nasamu A. S., Niles J. C. (2016) **Synthetic RNA-protein modules integrated with native translation mechanisms to control gene expression in malaria parasites** *Nat Commun* **7**
46. Filarsky M., et al. (2018) **GDV1 induces sexual commitment of malaria parasites by antagonizing HP1-dependent gene silencing** *Science* **359**:1259–1263
47. Yuda M., Kaneko I., Murata Y., Iwanaga S., Nishi T. (2021) **Mechanisms of triggering malaria gametocytogenesis by AP2-G** *Parasitol Int* **84**
48. Kent R. S., et al. (2018) **Inducible developmental reprogramming redefines commitment to sexual development in the malaria parasite Plasmodium berghei** *Nat Microbiol* **3**:1206–1213
49. Choi H., Kim S., Fermin D., Tsou C. C., Nesvizhskii A. I. (2015) **QPROT: Statistical method for testing differential expression using protein-level intensity data in label-free quantitative proteomics** *J Proteomics* **129**:121–126

50. Hoeijmakers W. A. M., et al. (2019) **Epigenetic reader complexes of the human malaria parasite, *Plasmodium falciparum*** *Nucleic Acids Res* **47**:11574–11588
51. Flueck C., et al. (2010) **A major role for the *Plasmodium falciparum* ApiAP2 protein PfSIP2 in chromosome end biology** *PLoS Pathog* **6**
52. Shang X., et al. (2022) **Genome-wide landscape of ApiAP2 transcription factors reveals a heterochromatin-associated regulatory network during *Plasmodium falciparum* blood-stage development** *Nucleic Acids Res* **50**:3413–3431
53. Chen Q., et al. (1998) **Developmental selection of var gene expression in *Plasmodium falciparum*** *Nature* **394**:392–395
54. Scherf A., et al. (1998) **Antigenic variation in malaria: in situ switching, relaxed and mutually exclusive transcription of var genes during intra-erythrocytic development in *Plasmodium falciparum*** *EMBO J* **17**:5418–5426
55. Meerstein-Kessel L., et al. (2018) **Probabilistic data integration identifies reliable gametocyte-specific proteins and transcripts in malaria parasites** *Sci Rep* **8**
56. Zhang M., et al. (2018) **Uncovering the essential genes of the human malaria parasite *Plasmodium falciparum* by saturation mutagenesis** *Science* **360**
57. Goldfless S. J., Wagner J. C., Niles J. C. (2014) **Versatile control of *Plasmodium falciparum* gene expression with an inducible protein-RNA interaction** *Nat Commun* **5**
58. Nasamu A. S., et al. (2021) **An integrated platform for genome engineering and gene expression perturbation in *Plasmodium falciparum*** *Sci Rep* **11**
59. Kiss T. (2002) **Small nucleolar RNAs: an abundant group of noncoding RNAs with diverse cellular functions** *Cell* **109**:145–148
60. Servant N., et al. (2015) **HiC-Pro: an optimized and flexible pipeline for Hi-C data processing** *Genome Biol* **16**
61. Bunnik E. M., et al. (2018) **Changes in genome organization of parasite-specific gene families during the *Plasmodium* transmission stages** *Nat Commun* **9**
62. Ardakany A. R., Ay F., Lonardi S. (2019) **Selfish: discovery of differential chromatin interactions via a self-similarity measure** *Bioinformatics* **35**:i145–i153
63. Varoquaux N., Ay F., Noble W. S., Vert J. P. (2014) **A statistical approach for inferring the 3D structure of the genome** *Bioinformatics* **30**:i26–33
64. Tencer A. H., et al. (2020) **Molecular mechanism of the MORC4 ATPase activation** *Nat Commun* **11**
65. Shao Y., et al. (2010) **Involvement of histone deacetylation in MORC2-mediated down-regulation of carbonic anhydrase IX** *Nucleic Acids Res* **38**:2813–2824
66. He F., et al. (2010) **Structural insight into the zinc finger CW domain as a histone modification reader** *Structure* **18**:1127–1139

67. Zhang Y., et al. (2019) **MORC3 Forms Nuclear Condensates through Phase Separation** *iScience* **17**:182–189
68. Luo J., Zeng S., Tian C. (2020) **MORC4 Promotes Chemoresistance of Luminal A/B Breast Cancer via STAT3-Mediated MID2 Upregulation** *Onco Targets Ther* **13**:6795–6803
69. Mimura Y., Takahashi K., Kawata K., Akazawa T., Inoue N. (2010) **Two-step colocalization of MORC3 with PML nuclear bodies** *J Cell Sci* **123**:2014–2024
70. Trager W., Jensen J. B. (2005) **Human malaria parasites in continuous culture. 1976** *J Parasitol* **91**:484–486
71. Fidock D. A., Nomura T., Wellems T. E. (1998) **Cycloguanil and its parent compound proguanil demonstrate distinct activities against Plasmodium falciparum malaria parasites transformed with human dihydrofolate reductase** *Mol Pharmacol* **54**:1140–1147
72. Rajaram K., Liu H. B., Prigge S. T. (2020) **Redesigned TetR-Aptamer System To Control Gene Expression in Plasmodium falciparum** *mSphere* **5**
73. Florens L., Washburn M. P. (2006) **Proteomic analysis by multidimensional protein identification technology** *Methods Mol Biol* **328**:159–175
74. Xu T., et al. (2015) **ProLuCID: An improved SEQUEST-like algorithm with enhanced sensitivity and specificity** *J Proteomics* **129**:16–24
75. Tabb D. L., McDonald W. H., Yates J. R. (2002) **3rd, DTASelect and Contrast: tools for assembling and comparing protein identifications from shotgun proteomics** *J Proteome Res* **1**:21–26
76. Zhang Y., Wen Z., Washburn M. P., Florens L. (2010) **Refinements to label free proteome quantitation: how to deal with peptides shared by multiple proteins** *Anal Chem* **82**:2272–2281
77. Choi H., Fermin D., Nesvizhskii A. I. (2008) **Significance analysis of spectral count data in label-free shotgun proteomics** *Mol Cell Proteomics* **7**:2373–2385
78. Zhang Y., et al. (2008) **Model-based analysis of ChIP-Seq (MACS)** *Genome Biol* **9**
79. Gupta M. K., Lenz T., Le Roch K. G. (2021) **Chromosomes Conformation Capture Coupled with Next-Generation Sequencing (Hi-C) in Plasmodium falciparum** *Methods Mol Biol* **2369**:15–25
80. Yang T., et al. (2017) **HiCRep: assessing the reproducibility of Hi-C data using a stratum-adjusted correlation coefficient** *Genome Res* **27**:1939–1949
81. Ay F., Bailey T. L., Noble W. S. (2014) **Statistical confidence estimation for Hi-C data reveals regulatory chromatin contacts** *Genome Res* **24**:999–1011
82. Goddard T. D., et al. (2018) **UCSF ChimeraX: Meeting modern challenges in visualization and analysis** *Protein Sci* **27**:14–25

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Reviewer #1 (Public review):

Summary:

The authors investigated the function of Microrchidia (MORC) proteins in the human malaria parasite *Plasmodium falciparum*. Recognizing MORC's implication in DNA compaction and gene silencing across diverse species, the study aimed to explore the influence of PfMORC on transcriptional regulation, life cycle progression and survival of the malaria parasite. Depletion of PfMORC leads to the collapse of heterochromatin and thus to the killing of the parasite. The potential regulatory role of PfMORC in the survival of the parasite suggests that it may be central to the development of new antimalarial strategies.

Strengths:

The application of the cutting-edge CRISPR/Cas9 genome editing tool, combined with other molecular and genomic approaches, provides a robust methodology. Comprehensive ChIP-seq experiments indicate PfMORC's interaction with sub-telomeric areas and genes tied to antigenic variation, suggesting its pivotal role in stage transition. The incorporation of Hi-C studies is noteworthy, enabling the visualization of changes in chromatin conformation in response to PfMORC knockdown.

Weaknesses:

Although disruption of PfMORC affects chromatin architecture and stage-specific gene expression, determining a direct cause-effect relationship requires further investigation. Furthermore, while numerous interacting partners have been identified, their validation is critical and understanding their role in directing MORC to its targets or in influencing the chromatin compaction activities of MORC is essential for further clarification. In addition, the authors should adjust their conclusions in the manuscript to more accurately represent the multifaceted functions of MORC in the parasite.

<https://doi.org/10.7554/eLife.92499.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public Review):

Summary: The authors investigated the function of Microrchidia (MORC) proteins in the human malaria parasite Plasmodium falciparum. Recognizing MORC's implication in DNA compaction and gene silencing across diverse species, the study aimed to explore the influence of PfMORC on transcriptional regulation, life cycle progression and survival of the malaria parasite. Depletion of PfMORC leads to the collapse of heterochromatin

and thus to the killing of the parasite. The potential regulatory role of PfMORC in the survival of the parasite suggests that it may be central to the development of new antimalarial strategies.

Strengths: The application of the cutting-edge CRISPR/Cas9 genome editing tool, combined with other molecular and genomic approaches, provides a robust methodology. Comprehensive ChIP-seq experiments indicate PfMORC's interaction with sub-telomeric areas and genes tied to antigenic variation, suggesting its pivotal role in stage transition. The incorporation of Hi-C studies is noteworthy, enabling the visualization of changes in chromatin conformation in response to PfMORC knockdown.

We greatly appreciate the overall positive feedback and cognisance of our efforts. Our application of CRISPR/Cas9 genome editing tools coupled with complementary cellular and functional approaches shed light on the importance of _Pf_MORC in maintaining chromatin structural integrity in the parasite and highlights this protein as a promising target for novel therapeutic intervention.

Weaknesses: Although disruption of PfMORC affects chromatin architecture and stage-specific gene expression, determining a direct cause-effect relationship requires further investigation.

Our conclusions were made on the basis of multiple, unbiased molecular and functional genomic assays that point to the relevance of the _Pf_MORC protein in maintaining the parasite's chromatin landscape. Although we do not claim to have precise evidence on the step-by-step pathway to which _Pf_MORC is involved, we bring forth first-hand evidence of its role in heterochromatin binding, gene-regulation and its association with major TFs as well as chromatin remodeling and modifying enzymes. We however agree with the comment regarding the lack of direct effects of _Pf_MORC KD and have since provided additional evidence by performing ChIP-seq experiments against H3K9me3 and H3K9ac during KD. Our new results are presented in Fig. 5. We showed that the level of H3K9me3 decreased significantly during _Pf_MORC KD.

Furthermore, while numerous interacting partners have been identified, their validation is critical and understanding their role in directing MORC to its targets or in influencing the chromatin compaction activities of MORC is essential for further clarification. In addition, the authors should adjust their conclusions in the manuscript to more accurately represent the multifaceted functions of MORC in the parasite.

Validation of the identified interacting partners is indeed critical and essential to understanding their role in directing MORC to its targets. Our protein pull down experiments have been done using several biological replicates. Several of the interacting partners have also been identified and published by other labs and collaborators. To confirm our results, we completed a direct comparison of our work with previous published work. Results have now been incorporated into the revised manuscript to confirm the identified interacting partners and the accuracy of the data we obtained in our experiment. Molecular validation of novel proteins identified in our protein pull down requires generation of tagged lines and may take a few more years but will be submitted for publication in a follow up manuscript.

Reviewer #2 (Public Review):

Summary: This paper, titled "Regulation of Chromatin Accessibility and Transcriptional Repression by PfMORC Protein in Plasmodium falciparum," delves into the PfMORC protein's role during the intra-erythrocytic cycle of the malaria parasite, P. falciparum. Le Roch et al. examined PfMORC's interactions with proteins, its genomic distribution in different parasite life stages (rings, trophozoites, schizonts), and the transcriptome's

response to PfMORC depletion. They conducted a chromatin conformation capture on PfMORC-depleted parasites and observed significant alterations. Furthermore, they demonstrated that PfMORC depletion is lethal to the parasite.

Strengths: This study significantly advances our understanding of PfMORC's role in establishing heterochromatin. The direct consequences of the PfMORC depletion are addressed using chromatin conformation capture.

We appreciate the Reviewer's comments and reflection on the importance of our work.

Weaknesses: The study only partially addressed the direct effects of PfMORC depletion on other heterochromatin markers.

Here again, we agree with the reviewer's comment and have performed additional experiments to delve deeper into the multifaceted roles of _Pf_MORC. We have performed additional ChIP-sequencing analysis on _Pf_MORC depleted conditions focusing on known heterochromatin and euchromatin markers H3K9me3 and H3K9ac respectively. We hope our new results presented in figure 5 will shed light on the more direct implications of _Pf_MORC on heterochromatin and gene silencing.

Reviewer #1 (Recommendations For The Authors):

Suggestions for improved or additional experiments, data or analyses.

• Why does MORC, which was used in the pull-down, seem to be only minimally enriched in the volcano plot, while a series of proteins (marked in red) and AP2 (highlighted in green) are enriched with log2 fold changes exceeding 15?

We apologize for the confusion. MORC was detected with the highest number of peptides (97 and 113) and spectra (1041 and 1177) confirming the efficiency of our pull-down. However, considering the relatively large size of the MORC protein (295kDa) and its weak detection in the control (5 and 7 peptides; 16 and 43 spectra), the Log2 FoldChange and Z-statistic after normalization are minimal compared to smaller proteins that were not identified in the control samples.

Additionally, can you explain why these proteins appear to be enriched at the same fold?

We can postulate that these proteins form a complex with a ratio of 1:1. Two of these three proteins are described to interact with MORC in several publications, supporting a strong interaction between them.

Variations in the interactome could result from the washing buffer's stringency.

We agree that the IP conditions could affect the detection of the interactome as well as the parasite stage used. As indicated below, the overlap with previous publications and the presence of AP2 TFs and chromatin remodelers strongly support our results.

It would be highly appropriate for the authors, similar to the co-submitted article (Maneesh Kumar Singh et al.), to present their mass spectrometry data in relation to previous purifications in Plasmodium (Bryant et al. 2020; Subudhi et al. 2023; Hillier et al. 2019) and also in Toxoplasma (Farhat et al. 2020). It would be good if authors could also put their results into perspective in light of the following pre-prints:

We agree with the reviewer's comment. In this revised manuscript, we compared our IP-MS data to previous published manuscripts. Key proteins including the AP2-P (PF3D7_1107800)

and HDAC1 were indeed identified in several experiments validating our initial findings of the formation of large complexes with MORC. However, it's important to highlight that the MORC protein was not used as the bait protein in previously published papers, and thus some discrepancies can be observed.

Given the tendency of MORCs to form multiple complexes with AP2 factors, have you explored whether specific AP2s are conserved between Plasmodium and Toxoplasma, within the phylum?

P. falciparum encodes for 27 putative AP2s, while *T. gondii* has over 60 AP2s, making direct comparison challenging. Some *Plasmodium* AP2s have multiple counterparts in *T. gondii* and typically conservation is limited to the AP2 binding domains. Attempts to identify sequence homology among AP2s and the regions of conservation have been performed (PMID: 30959972, PMID: 30959972, PMID: 16040597). Although this information would provide interesting insight, we believe exploring this topic at this time would diverge from our primary objectives. It would be more appropriate to address this in future studies.

Could this conservation be identified either through phylogenetic means or by using tools such as AlphaFold, especially considering not just the AP2 domains but also any existing ACDC domains?

Although this may reveal important information regarding the association between MORC proteins and AP2 domains, we believe investigating the conservation between AP2 across apicomplexan parasites may prove too challenging and is beyond the scope of this work.

Most of the genes are depicted without their immediate surroundings (Fig. 2d and Fig S2c, d). For instance, the promoter region of AP2g is not shown (Fig. 2d). It is therefore very challenging to determine the presence or absence of MORC upstream or downstream; considering that this factor, which can create DNA loop protrusions, might bind at a distance from the genes in question.

All gene coverage plots, including AP2-G, show 500 bp up- and downstream of the displayed gene. We have modified our figure legends to make sure that this information is provided.

Upon examining Figure S3, it is evident that the authors have indicated a decline in PfMORC expression, represented as percentages over two unique time frames. The methodology behind this quantification remains ambiguous. It's essential for the authors to specify whether normalization was done using a loading control. As a benchmark, Singh et al. (2021) in their Figure 4 transparently used GAPDH as a loading control and included an untreated sample in their western blot analysis.

We thank the Reviewer for bringing this to our attention. Our initial quantification was performed using ImageJ. To address the Reviewer's comment, we have reperformed the experiment. Our quantitative analysis was performed through Bio-Rad ImageLab software using aldolase expression as a loading control (50% of the MORC loading). This information has now been incorporated into the supplementary figures (Figure S3).

There's a striking observation that, despite significant degradation of PfMORC (as depicted in Figures S1 and S3), only the upper band in the western blot diminishes. This inconsistency needs addressing, as it can raise questions about the interpretation of the results.

We agree with the reviewer's comment. We experienced some challenges upon performing a Western Blot on such a large protein (295kDa). Our initial attempts required long exposure

that may have highlighted non-specific signals of smaller proteins. To address the reviewer's comment, we have performed the experiment one more time and made necessary changes to our WB protocol. Our new result better reflects the expected down regulation of _Pf_MORC. These changes have been incorporated to our manuscript and Fig S3.

Recommendations for improving the writing and presentation.

MORC KD quantification and consistency with previous findings (Figure S3): When comparing their results with those from another study (Singh et al. 2021), it's critical to ensure that the experimental conditions, especially the methodology for KD and the quantification of protein levels, are similar. If not, a direct comparison might be misleading.

We greatly appreciate the suggestions and have made efforts to redesign the MORC KD quantifications according to the reviewer's recommendations.

While the manuscript mentions the level of KD, it does not delve into the functional consequences of such a decrease in protein levels. It would be of interest to understand how this level of KD affects the parasite's biology, especially in the context of the paper's main findings.

We have addressed this question by looking at the changes in chromatin structure in WT versus KD parasites upon atc removal. We have also validated this initial result by designing an additional ChIP-seq experiment against histone marks in WT versus KD parasites upon atc removal. Our findings showed a significant downregulation in H3K9me coverage in heterochromatin regions, specifically in genes associated with antigenic variation and invasion genes. These findings suggest that PfMORC regulates at least partially gene silencing and chromatin arrangements. The manuscript has been edited accordingly.

Concluding page 5, the authors present an interpretation of their findings that suggests a multi-faceted role of PfMORC in regulating stage-specific gene families, particularly the gametocyte-related genes and merozoite surface proteins. While the narrative they present is intriguing, several concerns arise:

Over-reliance on correlation: The authors draw a direct line between the levels of PfMORC binding and the function of these genes in the parasite's life cycle. However, a mere correlation between PfMORC binding and stage-specific gene activity does not necessarily imply causation. They would need to provide experimental evidence showing that manipulation of PfMORC levels directly impacts these genes' expression.

We agree with the reviewer's comment. We have however partially addressed this issue by comparing our ChIP-seq, RNA-seq and Hi-C experiments. We concluded that several of the transcriptional changes observed were due to an indirect effect of PfMORC KD and were most likely induced by a cell cycle arrest and partial collapse of the chromatin structure. The collapse of the heterochromatin structure was validated using our Hi-C experiment. To further address additional concerns the review's had, we have included additional ChIP-seq experiments targeting histone marks to confirm our initial hypothesis. Result of this additional experiment has been incorporated in the revised version of the manuscript.

Ambiguity surrounding "low levels" and "high levels": The terms "low levels" and "high levels" of PfMORC binding are qualitative and could be subject to interpretation. Without quantification or a clear benchmark, these descriptions remain vague.

We agree with the reviewers that the terms "low levels" and "high levels" of PfMORC binding are qualitative and could be subject to interpretation. We have however quantified our

change in DNA binding using normalized reads (RPKM). In trophozoite and schizont stages, most of the genes contain a mean of <0.5 RPKM normalized reads per nucleotide of _Pf_MORC binding within their promoter region, whereas antigenic gene families such as *var* and *rifin* contain ~1.5 and 0.5 normalized reads, respectively (Fig. 2b). Similar results are also obtained for the gametocyte-specific transcription factor AP2-G that contains levels of _Pf_MORC binding similar to what is observed in *var* genes (Fig. 2c and S2c, d).

Shift in Binding Sites: The observed minor switch in PfMORC binding sites from gene bodies to intergenic and promoter regions is mentioned, but without context on how these shifts impact gene expression or any comparative analysis with other proteins showing similar shifts. The claim that this shift implicates PfMORC as an "insulator" is a leap without direct evidence.

We apologize for the confusion. We have compared our ChIP-seq with RNA seq results at different time points of the cell cycle and demonstrated that the shift observed has an effect in gene expression. We have edit the manuscript to clarify these results.

Overextension of PfMORC's Role: The authors suggest that PfMORC moves to the regulatory regions around the TSS to guide RNA Polymerase and transcription factors. This is a substantial claim and would require additional experiments to validate. Simply observing binding in a region is insufficient to assign a specific functional role, especially one as critical as guiding RNA Polymerase. Historically, the MORC family has been primarily linked with gene silencing across Apicomplexan, plants, and metazoans. On page 7, the authors noted a minimal overlap between the ChIP-seq and RNA-seq signals (Fig. 4e). They also acknowledged that the pronounced gene expression shifts at schizont stages result from a combination of direct and indirect impacts of PfMORC degradation, which could cause cell cycle arrest and potential heterochromatin disintegration, rather than just decreased PfMORC binding. Therefore, the authors should adjust their conclusions in the manuscript to more accurately represent the multifaceted functions of MORC in the parasite.

We agree with the reviewer's comment and have edited the manuscript accordingly.

DISCUSSION:

The authors concluded that "Using a combination of ChIP-seq, protein knock down, RNA-seq and Hi-C experiments, we have demonstrated that the MORC protein is essential for the tight regulation of gene expression through chromatin compaction, preventing access to gene promoters from TFs and the general transcriptional machinery in a stage specific manner."

Again, the assertion that MORC protein is essential for tight regulation of gene expression, based purely on correlational data (e.g., ChIP-seq showing binding doesn't prove functionality), assumes causality which might not be fully substantiated. The phrase "preventing access to gene promoters from TFs and the general transcriptional machinery in a stage-specific manner" needs also validation. Asserting that MORC is essential for this function might oversimplify the process and overlook other critical contributors.

We agree with the reviewer's comments and the conclusion has since been edited accordingly.

The discussion is quite poor. It would be pertinent to put MORC in perspective within the broader picture of regulatory mechanisms of chromatin state at telomeres and var

genes. For instance, how do SIR2 and HDAC1 (associated with MORC) divide the task of deacetylation? Or the contribution of HP1 and other non-coding RNAs.

We agree with the reviewer's suggestion. However, in order to put MORC in perspective within a broader picture, we would need to measure changes in localization of several molecular components regulating heterochromatin in WT versus KD condition. This will require access to several molecular tools and specific antibodies that we do not currently have. We have addressed these issues in our discussion.

Minor corrections to the text and figures.

Figure 1d: Could you provide the ID for each AP2 directly on the volcano plot? While some IDs are referenced in the manuscript, visual representation in the plot would facilitate a clearer understanding of their enrichment levels.

ID for unknown AP2 proteins have been added on the volcano plot.

I recommend presenting Figure S2b as a panel within a primary figure. This change would offer readers a more quantitative understanding of the distinct differences between developmental stages. Notably, there seems to be a limited number of genes in common when considering the total, and there is an apparent lack of enrichment in the ring stage.

This has been done.

The captions are very minimally detailed. An effort must be made to better describe the panels as well as which statistical tests were used.

We have improved the figure legends and add the number of biological replicates as well as the statistic used in each figure legend.

Figure 1A: The protein diagram with its domains does not take scale into account.

The figure has been modified.

Reviewer #2 (Recommendations For The Authors):

(1) The study lacks a direct link between PfMORC's inferred function and the state of heterochromatin in the genome post-depletion.

We agree with the reviewer's comment and have included additional ChIP-seq experiments to measure changes in histone marks in PfMORC depleted parasite line. We show a significant decrease in histone H3K9me3 marks in PfMORC KD condition.

Conducting ChIP-seq on well-known heterochromatin markers such as H3K9me3, HP1, or H3K36me2/3 could shed light on the consequences of PfMORC depletion on global heterochromatin and its boundaries.

With no access to an anti-HP1 antibody with reasonable affinity, we have not been able to study the impact of MORC KD on HP1 but have successfully observed the impact on H3K9me3 marks. These results have been added to the revised manuscript in (Fig. 5).

(2) The authors should conduct a more comprehensive analysis of PfMORC's genomic localization, comparing it to ApiAP2 binding (interacting proteins) and histone modifications. This would provide valuable insights.

We have performed a more comprehensive genome wide analysis of MORC binding through ChIP-seq on WT and MORC-KD conditions. Our results show that _Pf_MORC localizes to heterochromatin with significant overlap with H3K9-trimethylation (H3K9me3) marks, at or near *var* gene regions. When downregulated, level of H3K9me3 was detected at a lower level, validating a possible role of _Pf_MORC in gene repression. Regarding the comparison with AP2 binding, our proteomics datasets have shown extensive MORC binding with several AP2 proteins.

(3) RNA-seq data reveals that only a few genes are affected after 24 hours of PfMORC depletion, with an equivalent number of up-regulated and down-regulated genes. The reasons behind down-regulation resulting from a heterochromatin marker depletion are not clearly established.

We agree with the reviewer's comment. At this stage (24 hours), _Pf_MORC depletion is limited and the effects at the transcriptional level are quite restricted. Furthermore, it is highly probable that down-regulated genes are most likely due to an indirect effect of a cell cycle arrest. We have edited the manuscript to address this comment.

The relationship between this data and the partial depletion of PfMORC needs further discussion.

We agree with the reviewers and have improved our discussion in the revised version of the manuscript.

(4) The authors did not compare their ChIP-seq data with the genes found downregulated in the RNA-seq data. Examining the correlation between these datasets would enhance the study.

We apologize for the confusion. We have compared ChIP-seq and RNA-seq data and identified a very limited number of overlapping genes indicating that most of the changes observed in gene expression are in fact most likely indirect due to a cell cycle arrest and a collapse of the chromatin. We have edited the manuscript to clarify this issue.

(5) The discussion section is relatively concise and does not fully address the complexity of the data, warranting further exploration.

We have improved the discussion section in the revised version of the manuscript.

<https://doi.org/10.7554/eLife.92499.2.sa0>